

When Better APT Scores Hide Missed Attack Warnings

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Abstract of Praxis

When Better APT Scores Hide Missed Attack Warnings

This praxis examines when improved advanced persistent threat classification scores accompany missed attack warnings. It contributes controlled evidence and an executable procedure reporting exact-stage recognition, attacks classified as benign, and benign false alerts together. In a UNRAVELED acquisition comparison, mean macro-F1 increased from 0.7148 to 0.7379 while exfiltration warning recall decreased from 85.18% to 76.25%; omission analyses revealed substantial effect-size sensitivity. A fixed-anchor temporal contrast increased macro-F1 by 0.0632 when later-period observations entered fitting. A full-release extension verifies 173 files and 6,877,157 rows and fits current-flow and role-augmented models without class caps in six sensor views. Roles improve macro-F1 in two views, while stage support constrains interpretation. An adapted external-source test additionally inspects all 3,465,342 AIT netflow rows, trains on six executions and evaluates two later executions containing 1,067,211 eligible rows. Its three-class, history-only contract differs from the original four-class, two-group task. The original error-focused-versus-entropy score-up/warnings-down direction recurs in none of six execution-by-seed comparisons. Wilson shows a small opposite-order tradeoff: higher F1 under entropy accompanies one to eight additional missed exfiltration warnings; Harrison improves both outcomes. The AIT computational audit passes 1,245 checks. These results support joint reporting while limiting claims about policy superiority and recurrence. Shared laboratory scenarios, publisher-defined labels and completed-flow features leave unrelated-campaign generalization, early forecasting and operational benefits unresolved.

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List of Symbols

x : observed feature vector

y : declared evaluation class

p : model class-score vector

g : fitted history selector

w : declared stage-cost vector

B : simulated evidence budget

D : simulated decision deadline

$C(s,j)$: confusion count for true class s and predicted class j

List of Acronyms

APT: Advanced Persistent Threat

AUC: Area Under the Curve

EDA: Exploratory Data Analysis

GML: Graph Machine Learning

GMR: Graphical Model of Research

IDS: Intrusion Detection System

ML: Machine Learning

ROC: Receiver Operating Characteristic

RQ: Research Question

Chapter 1: Introduction

1.1 Background

A security analyst reviewing a suspicious transfer faces two related questions: should this activity receive an attack warning, and which attack stage best describes it? A system can answer the second question incorrectly while still drawing attention to the activity. It can also assign a benign label and remove that opportunity for review. These outcomes have different operational meanings, even when a multiclass objective assigns them the same penalty.

Consider 100 records labeled exfiltration by a dataset author. If a model calls 60 exfiltration, 20 another attack stage, and 20 benign, its exact-stage recall is 60% and its warning recall is 80%. This illustrative accounting is not a study result. It explains why a single headline score cannot describe both stage accuracy and preservation of an attack label. Warning recall must also be considered with benign false alerts: predicting an attack for every record preserves every warning and creates an unusable workload.

Uddin et al. (2025) provide a direct empirical precedent for this distinction. Their hierarchical-versus-flat intrusion study reports per-attack recognition and attacks misclassified as normal. Recent APT provenance work also challenges the relationship between conventional evaluation scores and practical detection behavior (Bilot et al., 2025; Guerra et al., 2026). These findings motivate closer measurement of the decisions represented by a score, rather than assuming that a new model architecture solves the evaluation problem.

1.2 Research Motivation

Security-model review is an engineering decision: a practitioner must decide whether a proposed change improves outcomes that matter for the intended use. A multiclass score summarizes several kinds of error, yet those errors do not have the same meaning for an investigation. An event labelled as the wrong attack stage can still receive attention; an event labelled benign may receive none. Historical context can change both the stage label and the error destination.

This study began with two practical interventions: learning when historical context should be used and choosing additional evidence under a collection budget. Their completed comparisons revealed a useful evaluation question. Improvements in aggregate classification performance sometimes occurred alongside a loss of attack warnings. The praxis therefore centres on measuring that relationship and providing an inspection procedure supported by the recorded comparisons. The model implementations serve as experimental tools for this applied question.

1.3 Problem Statement

Problem statement. APT-stage evaluation can reward a change in aggregate classification performance without making its effects on missed attack warnings and benign alert workload visible, while available benchmark artifacts may not support the temporal comparison needed to interpret that change.

The purpose of this praxis is to measure these effects under explicit controls and provide an executable audit that practitioners can apply before accepting a stage classifier. The immediate application is model evaluation and selection. It is not a production detector deployment or a forecast of data theft.

The industry consequence is concrete: a team comparing model updates could accept a higher F1 score while sending fewer exfiltration-labeled events for investigation. A team can also overestimate its evidence by counting many correlated flows as independent incidents or by treating related dataset releases as separate replications. The study measures examples of these issues and documents the source checks needed to interpret them.

1.4 Thesis Statement

Evaluating APT models using overall performance, stage-specific missed warnings and benign false alerts together reveals consequential tradeoffs that an overall score alone does not describe. A defensible comparison must also establish that its training chronology, evaluation population and source labels support the question being asked.

The thesis is evaluated as an empirical measurement contribution. It does not require a new detector architecture. Its evidence comprises controlled comparisons, observed error destinations, sensitivity analyses and an executable source-qualification procedure.

1.5 Research Objectives

The objectives are to:

1. Quantify how historical-evidence choices change macro-F1, exact-stage recognition, stage-conditioned warning recall and benign false alerts on the same evaluation records.

2. Measure training-composition sensitivity while holding the classifier family, evaluation anchor and class-specific fitting budgets fixed.
3. Determine whether four proposed APT releases can support the declared all-native-class chronological comparison.
4. Deliver traceable results, understandable model mathematics and a reproducible reporting procedure that keeps adverse outcomes and uncertainty visible.

1.6 Research Questions and Analytical Propositions

RQ1, primary: When historical-evidence choices improve an APT model's overall score, what happens to exfiltration warnings and false alarms on the same evaluation records?

RQ2, supporting: How much does access to later-period fitting observations change reported stage performance when architecture, evaluation records and per-class fitting counts remain fixed?

RQ3, supporting: Which proposed benchmark artifacts support the unchanged all-native-class chronological comparison, and what prevents the others from supporting it?

The evaluation proposition for RQ1 is that a higher macro-F1 does not ensure preservation of warnings for a particular attack stage. For RQ2, the tested expectation is that changing temporal access and training composition can change scores despite a fixed evaluation anchor. For RQ3, the necessary-condition proposition is that a single cutoff cannot provide the specified earlier and later native-class support when the class-support interval is empty.

The fitted experiments froze their protocols before fitting. The warning-loss pattern was inspected retrospectively; these propositions do not retroactively constitute newly preregistered hypotheses. The complete comparisons and unfavorable outcomes remain in the results. This edition places the warning question first for clarity; the original experiment identifiers, protocols and results retain their original labels in the evidence archive.

1.7 Scope of Research

The main study evaluates the prepared UNRAVELED flow artifact and uses its four declared evaluation classes. The history-selection, evidence-acquisition and temporal experiments contribute 141 fitted models; a separate two-fit policy-transfer supplement brings the wider batch to 143. These fitting counts are not counts of independent attacks.

The September 27 full-release extension additionally audits all 173 released flow files and all 6,877,157 source rows across six sensor views, fitting current-flow and current-flow-plus-role models on every eligible training row. Its 12 fitting jobs bring the combined recorded inventory to 155; single-class training jobs are explicitly identified as degenerate controls. The new extension extends population coverage and removes fitting caps for the role-feature comparison. It does not rerun every historical intervention on the expanded population and does not create a second campaign.

The source-qualification study covers inspected SCVIC-APT-2021, DAPT2020 and DSRL-APT-2023 artifacts and the acquisition status of S-DAPT-2026. It does not claim four additional fitted replications. CasinoLimit and CAM-LDS support a separately labelled technique-recognition supplement, whose units and negative labels differ from the main benign-versus-attack stage task.

The September 28 external-source extension qualifies all eight AIT netflow testbeds and evaluates a separately frozen three-class, one-history-group configuration on two later, held-out executions. It inspects 3,465,342 source rows and uses every eligible row: 2,397,158 for final training and 1,067,211 for evaluation. Its 36 classifier and six selector fits bring the recorded project inventory to 197 fitting jobs, including earlier constant-class controls. Separate source executions broaden the empirical record; the repeated laboratory scenario, different class taxonomy and different evidence groups limit the test to an adapted validation. Sections 3.12 and 4.11 state those changes before interpreting its results.

1.8 Research Limitations

The main results describe one previously examined campaign. Its movement target denotes author-annotated remote discovery on one host pair, with only 35 evaluation rows and 18 anchor rows. Completed-flow features do not measure detection before flow completion. Stage labels do not independently verify successful compromise or delivery of stolen files. Simulated missing, delayed and wrong-host evidence probes specified conditions; it does not estimate their frequency or price in production.

The paired warning analysis is retrospective, and the five capture fragments are correlated parts of a shared workflow. Conditional resampling intervals and fitting-seed sensitivity describe these data rather than a population of independent campaigns. Chapter 5 states the implications of these limits for the empirical claim.

The added AIT evaluation uses two held-out executions from a common laboratory generator. It supplies an external-source test for the adapted configuration, while leaving the original four-class, two-group experiment and real-world generalization

distinct. The frozen original-direction test is unreplicated; a much smaller warning tradeoff appears in one execution with the opposite policy ordering.

1.9 Praxis Organization

Chapter 2 reviews the relevant literature and identifies the narrow empirical contribution. Chapter 3 presents the Graphical Model of Research, source evaluation, implemented models, mathematical objectives and evaluation controls. Chapter 4 reports the completed results and sensitivity analyses. Chapter 5 discusses their meaning, contributions, limitations and recommendations. The appendices and accompanying machine-readable supplement retain comprehensive result coverage, provenance and the original GML GMR reference.

Chapter 2: Literature Review

2.1 Introduction

This review positions the study relative to recent APT evaluation, temporal validity and error-destination research. It is a targeted primary-source review completed September 23, 2026, centred on 2024-2026 publications and supplemented with original dataset and method sources. It identifies overlap and concrete distinctions; it is not a systematic review or an exhaustive priority claim.

2.2 Temporal Validity in Security Evaluation

TESSERACT established temporal and distributional constraints for evaluating malware classifiers and showed why inappropriate splits can produce misleading conclusions (Pendlebury et al., 2019). This is foundational prior art rather than part of the recent-literature window. Holding out random records is not equivalent to asking whether a model trained earlier will work later.

Bilot et al. (2025) examine provenance-based intrusion detectors in a common framework, identify practical and evaluation shortcomings, and include simple alternatives. Their work supports the use of strong simple controls and detection-relevant outcomes. Guerra et al. (2026) directly address APT provenance benchmarking, including temporally separated evaluation and the influence of benchmark semantics on conclusions. Consequently, neither temporal hygiene nor critical measurement of APT detection is new in itself.

The present temporal contrast is narrower: it keeps the later evaluation rows and per-class fitting budget constant while changing access to later-period training observations. This avoids attributing a difference caused by a different test population to

training chronology alone. It still changes training composition and diversity; it cannot identify an effect of time independently of every other property of those added observations.

2.3 Correct Attack Stages and Retained Warnings

Uddin et al. (2025) compare hierarchical and flat intrusion classifiers across ten datasets and ten algorithms. The available author manuscript (Uddin et al., 2024) reports exact attack recognition, attacks classified as normal, confusion matrices, and false-positive tradeoffs. The distinction between a wrong attack type and a missed attack therefore has direct prior coverage. Renaming stage-conditioned binary recall would not create a new metric.

The empirical question addressed here is whether a fixed experimental comparison can improve macro-F1 while reducing warnings for an author-defined consequential stage. The answer must include benign workload and error destinations, because a warning-retention objective alone can be satisfied trivially. The paper reports this tradeoff from saved predictions and preserves its retrospective discovery status.

2.4 Recent Flow and Attack-Stage Research

The 2026 TAN-IDS framework provides a deployment-oriented, shared NetFlow evaluation interface with in-domain and cross-domain comparisons (Ha Thanh, 2026). Its stated limitations leave multiclass or family discrimination and systematic feature ablation for further work. This is a concrete scope boundary that motivates examining stage-specific errors under controlled changes in data and evidence.

Recent APT research already investigates temporal and contextual attack-stage recognition. StageFinder combines structural and temporal information for stage

estimation (Phan & Bauschert, 2026). Other contemporary flow studies and dataset efforts investigate sequence or graph representations. The current work therefore does not claim that temporal context, ensembles, or flow-based APT stages are unexplored. It evaluates how particular contextual decisions affect a useful warning and how much the evaluation design contributes to the reported score.

Recent model comparisons and flow-sequence studies include Luengo Viñuela et al. (2026), Iturbe et al. (2026), and Ibrahim et al. (2025). SANGl also examines sequential network patterns and graph learning for APT detection (M K et al., 2026). These are direct application precedents; their published scores are not reproduced baselines in this paper. Bibliographic name forms follow the publishers' records, with source details retained in the reference audit.

Othman et al. (2026) study DAPT stage durations and residual time-to-compromise through a survival-modelling task. Their session-based timing question differs from requiring every native class in both earlier training and later testing for a closed-set classifier. The support limitation measured here applies to the latter requirement and does not invalidate the former use of DAPT.

2.5 Dataset Origin and Benchmark Independence

UNRAVELED is a semi-synthetic APT dataset published in 2023 (Myneni et al., 2023). Its use here is necessary data provenance, not a claim to use a newly released 2026 corpus. The proposed expansion includes SCVIC-APT-2021 (Liu et al., 2022a, 2022b), DAPT2020 (Myneni et al., 2020), DSRL-APT-2023 (Shadabfar et al., 2025), and S-DAPT-2026 (Tijjani et al., 2026, withdrawn). Release names and paper dates do not establish independent executions, valid event clocks, or accessible source bytes.

DSRL is especially relevant to independence: its paper describes synthetic attacks generated from DAPT and benign examples sampled from DAPT. These relationships must be retained in any evidence count. Likewise, the inspected S-DAPT arXiv record is withdrawn, and the present project did not acquire a qualified replacement data release. An unavailable artifact cannot become a performance result through its citation alone.

The AIT netflow release (Soro et al., 2024) derives TCP and UDP flow records from eight executions of the AIT laboratory environment. Its simulated legitimate activity and attack scenarios are described by Landauer et al. (2023), building on the model-driven testbed approach of Landauer et al. (2021). The released flow labels support benign activity, other attacks and DNS exfiltration. They do not supply the same movement class or evidence channels as the original UNRAVELED experiment. This distinction motivates the explicitly adapted validation in Section 3.12, rather than treating a dataset name as proof of an unchanged replication.

2.6 Literature Gap and Contribution Positioning

The working gap is the lack of the specific controlled evidence assembled here: fixed-anchor training-composition effects, joint accounting of stage errors and lost warnings under historical-evidence choices, and an executable qualification audit of the requested APT-flow releases. Close prior work covers each surrounding concept. The claim is an applied measurement contribution with explicit experimental controls and artifacts, not proof that no earlier paper contains any related observation.

The literature review is targeted, not systematic. It checks named closest work and recent primary sources on temporal APT evaluation, hierarchical error destinations, stage reasoning, and source datasets. The reference and claim audit records publication status,

accessible sections, metadata, and limits of verification. The accessible 2024 author version of the Uddin study is distinguished from its 2025 journal record; available 2026 preprints are distinguished from proceedings not yet published.

Table 2-1. Closest literature and study scope

Closest work	Established contribution	Scope of this study
TESSERACT (2019)	Temporal and distribution constraints in malware evaluation	Same-anchor APT-flow training-pool contrast
Bilot et al. (2025); Guerra et al. (2026)	Critical provenance/ APT benchmarking and practical controls	Flow-stage observations with explicit source support
Uddin et al. (2025)	Attack-type errors versus attacks called normal	Measured metric/warning tradeoffs under evidence changes
TAN-IDS (2026)	Flow-based cross-domain evaluation; binary scope	Stage-conditioned warning outcomes and temporal controls
Othman et al. (2026)	DAPT stage timing and residual-time survival analysis	Necessary support for a different, all-class classifier split

2.7 Machine-Learning Foundations and Their Role

The main experiments use gradient-boosted decision trees for multiclass prediction and scalar regression. Boosting adds trees sequentially to improve a specified loss; using the same family across comparisons limits architecture changes as an explanation for a score difference. Chapter 3 distinguishes the multiclass experts from the regressors that select history or additional evidence, and reports their actual objectives.

The separate policy-transfer supplement uses previously fitted logistic-regression experts and newly fitted Ridge selectors. These models provide a deliberately modest test of whether a learned context-selection score changes the decisions of its source experts. The model-specific primary references and equations are given alongside their

implementations in Chapter 3. No graph-neural model, TabM architecture or large language model was fitted in this completed measurement batch.

2.8 Literature Review Summary

The literature establishes that time ordering, source semantics and error destinations matter in security evaluation. The empirical contribution here is a controlled flow-level account of their interaction: fixed-anchor training-composition comparisons, stage-conditioned warning and workload measurements under evidence choices, and a task-specific audit of named dataset releases. Existing metric definitions are used directly. The purpose is to add inspectable measurements and a reusable applied procedure, while keeping the distinction between published precedent and this study's observed results clear.

Chapter 3: Methodology

3.1 Introduction and Graphical Model of Research

The Graphical Model of Research (GMR) summarizes the completed workflow. It follows the numbered What/Why/How convention in the author's earlier GWU GML praxis (Pagan, 2026), with the content adapted to the present measurement study. The original graph-construction, architecture-training and tuning diagram is retained separately as a historical reference in Appendix C.



Figure 3-1. Graphical Model of Research for the completed APT evaluation study. The six stages connect source qualification, controlled fitting, error destinations, sensitivity checks and a reproducible evidence package.

The first two stages establish which records, targets and chronology are available.

The third fits the declared comparison models without selecting a new architecture from

evaluation scores. The fourth records what each prediction does to the attack warning and the benign workload. The fifth evaluates conditional uncertainty and source support. The sixth assembles the completed evidence into a reproducible praxis. The figure is a workflow description, not an additional empirical result.

3.2 Research Design and Analysis Chronology

The study combines completed controlled experiments with a retrospective analysis of their saved predictions and a separate benchmark-qualification study. These components answer different questions and have different evidentiary status.

The history-selection, evidence-acquisition, and temporal experiments each froze protocol and executable source before their fitting runs. The source corpus and portions of the broader task had already been examined in project development. The first warning-loss observation was added after inspecting an acquisition seed; it remains exploratory. The new paired reanalysis freezes its computation before execution, but neither that freeze nor its intervals convert an observed pattern into a prospectively confirmed hypothesis.

The D1 extension separately registers future comparisons and source eligibility requirements. Qualification and support checks completed before any new model fits. Those checks are reported as results. No model comparison is reported for a source that failed its applicable requirements.

Table 3-1. Components of the evidence and their roles.

Component	Completed scope	Role in this paper
History selection	39 fits; 7 arms; 5 conditions; 3 seeds	Context tradeoffs and simple controls
Evidence acquisition	84 fits; 3 budgets; 3 conditions; 3 seeds	Paired stage/warning outcomes
Temporal comparison	18 fits; 2 feature views; 3 split arms; 3 seeds	Fixed-anchor protocol sensitivity
Paired reanalysis	36 comparisons; no new fits	Retrospective, capture-conditional uncertainty
Benchmark qualification	4 requested sources; no new fits	Native support, timing and dependencies
Full-release extension	173 files; 6,877,157 rows; 12 fitting jobs	Uncapped current-flow versus role-feature comparison; same campaign
Policy-transfer supplement	2 fits; T1105 recognition	Separate task; not stage replication
AIT external-source validation	8 executions; 3,465,342 rows; 42 fits	Three-class history acquisition; 2 later executions held out

The directional expectations were that later-period training may increase reported performance, that context or evidence selection may change stage and warning outcomes differently, and that some releases may not support an all-stage temporal comparison. Results in either direction are informative. No 90% recall requirement or minimum favorable effect determines whether an observed finding is retained.

3.3 Datasets, Evaluation Support and Feature Interpretation

3.3.1 Why source qualification is part of the experiment

An APT dataset name does not establish an independent attack execution, a valid clock, or a label for successful movement or theft. This study therefore separates datasets used for completed predictions from artifacts inspected for possible extension. The

qualification decisions below describe the inspected releases and the registered comparison, not the usefulness of each dataset for every research question. Source-status statements are inherited from the verified September 23, 2026 audit; these historical qualification outcomes are preserved. The full-release extension in Section 3.11 additionally verifies all UNRAVELED flow-file bytes and performs new fitting.

Table 3-2. Dataset roles and qualification boundaries

Dataset	Role in this paper	Inspected support	Interpretation boundary
UNRAVELED	Historical controlled experiments	382,229 prepared flows; eleven IT-sensor captures	One previously examined campaign; author stages
SCVIC-APT-2021	Extension qualification	259,120 rows; six native classes	Recorded timestamps do not establish physical chronology
DAPT2020	Extension qualification	86,691 rows; five native classes	No all-native-class single chronological cutoff under the stated rule
DSRL-APT-2023	Extension qualification	65,000 rows; five native classes	Synthetic attacks and reused benign rows derived from DAPT
S-DAPT-2026	Source-access qualification	No qualified source bytes	No measured dataset count or model result
CasinoLimit	Secondary T1105 policy development	Evaluation: 920 targets, 17 T1105 positives	Annotation-onset proxies; negatives are other techniques
CAM-LDS	Secondary T1105 policy transfer	Evaluation: 4,209 targets, 100 T1105 positives	Labeled interval proxies; distinct decision unit

3.3.2 Primary corpus: UNRAVELED

UNRAVELED is the semi-synthetic APT corpus of Myneni et al. (2023), published in *Computer Networks*. The pinned author release is the UNRAVELED repository, commit d2ea90055d82fa448ab20588a13e3ec8bfd74816. The upstream

inventory records 173 network-flow files, 31 capture directories, and 6,877,157 rows.

The historical controlled experiments use a fixed subset of eleven complete locally available net1013x IT-sensor capture files, not all source traffic and not eleven independent attacks. Using a single sensor avoids indiscriminately pooling overlapping gateway/subnet observations. It does not establish complete visibility of the network.

The 382,229-row prepared artifact is bound by SHA-256 b2a491474e722f4dabcd4c419c83a4a6b49f08dfc3bc059aa42ef2aaa4c3de14. It was already examined in prior development. Captures 0-4 provide earlier fitting data, capture 5 is the original calibration partition, and captures 6-10 provide later evaluation. All history entries finish before the corresponding current flow begins. Current features describe completed flows, so the model decision is not an early forecast.

Table 3-3. UNRAVELED class support by partition

Partition	Benign	Other attack stage	Movement label	Exfiltration label
Earlier fitting pool	147,087	12,362	27	1,740
Original calibration	8,929	2,659	0	1,331
Later evaluation	192,193	12,424	35	3,442
Fixed temporal anchor, a subset of later evaluation	96,098	6,213	18	1,722

The original calibration partition contains no movement examples and cannot calibrate movement recall. The fixed temporal anchor totals 104,051 rows. The temporal study's conventional random comparison has a different 210,226-row population, including 191,515 benign, 15,095 other-stage, 34 movement, and 3,582 exfiltration labels. These denominators must not be substituted for the fixed-anchor denominators.

3.3.3 Source parsing and target mapping

The upstream preparation found unquoted commas in descriptive DPI fields of the source CSVs. It recovered the four annotation fields from the right end of each row, preserving the stable numeric prefix and excluding the ambiguous text region. This correction preceded the studied fits. A fixed-position parser would have interpreted some descriptive values as spurious stages. The raw source files and prior frozen artifacts were not rewritten.

The models use a declared four-class target: Benign, OtherAttackStage, LateralMovement, and DataExfiltration. The mapping combines source Reconnaissance, Establish Foothold, and Cover up labels into OtherAttackStage; it does not claim that the source has only four native stages. In the selected eleven captures, the other-stage rows consist of foothold and cover-up annotations. The mapping and full native-source inventory remain in the upstream qualification artifacts.

For this sensor, the movement activity is **Remote System Discovery on one directed host pair**. Those labels do not establish successful remote login, compromise, or movement to a new host. The exfiltration class is also the author's stage annotation, not independently verified stolen-file receipt. Annotation fields, including Stage, Activity, DefenderResponse, and Signature, are prohibited as predictors. In particular, a DefenderResponse value of Benign is not the traffic's true benign label.

3.3.4 Prepared feature groups

Table 3-4. Prepared features and availability

Feature group	Actual representation	Availability and excluded information
Current completed flow	Numeric duration, packet/byte counts, packet-size and inter-arrival summaries, TCP flag counts; three destination-service indicators	Full-flow measurements; no literal host identities, absolute dates, capture names, or annotation columns
Coarse endpoint roles	Eight one-hot values: four source categories and four destination categories	Department, public services, private services, other address; derived from documented static topology
Earlier activity	Thirty-six summaries: nine state values $\times$ source/destination $\times$ five/thirty-minute windows	Only previously completed flows, ending strictly before current-flow start
Selector availability signals	Relative history age and declared availability indicators where included in the protocol	Observable signals; no true stage provided to the deployed selector

The nine history values are logged completed-flow count, transmitted bytes, received bytes, distinct peers, initiated flows, remote-administration flows, internal-peer flows, whether the current peer appeared in the window, and whether the host had appeared previously. Numerical counts use the source's $\log_{1p}$ transformation. Other address does not automatically mean public Internet, and coarse role is not a verified per-user or per-machine business function. These definitions come from the existing feature preparation and history implementation.

The acquisition experiment treats precomputed role and history summaries as hypothetical information requests. A role lookup costs one simulated unit; history costs two. Delivery and failure schedules are shared across compared policies. This is a test of acquisition decisions under declared assumptions, not measurement of actual sensor costs or latency. Missing, stale, and wrong-host histories are explicit synthetic interventions.

3.3.5 A real published aggregate example

No previously published individual feature-value row was found in the inspected public evidence. An individual flow is therefore not fabricated or presented as though it were observed. The documented schema above shows the actual prepared fields; the following example is an **actual aggregate evidence record**, with endpoint identities absent.

Table 3-5. Published aggregate example

Field in saved clean evaluation, seed 20260924	Recorded value
Model arm	Current flow plus roles
Evaluation capture group	6
Total flow records	70,451
True-class counts: benign / other / movement / exfiltration	62,400 / 5,780 / 18 / 2,253
Movement label predictions: benign / other / movement / exfiltration	3 / 0 / 15 / 0

This example is copied from the source aggregate in PX080 and can be independently checked in the bound source snapshot. It illustrates both the denominator and the distinction between exact-stage recognition and a missed warning. It contains no invented feature values or assertion that the eighteen labels represent eighteen independent intrusions.

3.3.6 Extension artifacts and native stage support

The native-class support chart is reported with the qualification results in Chapter 4.

3.3.7 SCVIC-APT-2021

Liu et al. (2022a) published the benchmark in *IEEE Networking Letters*, with a separate IEEE DataPort dataset record. The inspected training CSV contains 254,836 NormalTraffic, 73 InitialCompromise, 833 Reconnaissance, 729 LateralMovement, 2,122 Pivoting, and 527 DataExfiltration rows. All six classes remain distinct in qualification.

A recorded-time ordering can be computed, and the necessary all-class start-time support interval is nonempty. Physical chronology remains unresolved: 220 benign rows parse to January 17, 1970; the other rows parse to October 21, 2015; timestamps mix minute and second resolution; and no qualified row-to-execution mapping or author-supported clock correction was found. A cached source page lists a separate test artifact, but its contents and independence were not qualified in the inspected workspace. These findings permit a carefully scoped random/grouped development question, not an unchanged deployment-valid temporal claim.

3.3.8 DAPT2020

Myneni et al. (2020) describe DAPT2020 as a benchmark for advanced persistent threats. The inspected ten CSVs cover five capture dates and contain 63,712 benign, 11,909 reconnaissance, 8,604 foothold, 2,451 movement, and fifteen exfiltration labels. Ten files are collection units within an attack progression, not ten campaigns.

The necessary single-cutoff support test requires at least two earlier fitting rows and one later evaluation row for every native class. The cutoff must exceed each class's second-earliest start and be no later than every class's latest start. The resulting lower bound is July 19, 2019 at 16:38:37, imposed by exfiltration; the upper bound is July 17 at 19:24:55, imposed by reconnaissance. The interval is empty. Thus no single cutoff

satisfies that stated all-class rule on these bytes. Completed-flow availability and duplicate controls can only make the rule more restrictive.

This is a precise support result, not a failed detector or a judgment that DAPT is unusable. Different questions, such as unknown-stage recognition or stage-duration/survival analysis, require different protocols. The benchmark can remain useful for those tasks without meeting this paper's closed-set chronological requirement.

3.3.9 DSRL-APT-2023

Shadabfar et al. (2025) describe DSRL as a synthetic dataset. The inspected file has 10,000 benign, 14,366 reconnaissance, 22,968 foothold, 12,664 movement, and 5,002 exfiltration rows. The publisher paper's Section 5 states that the attack generation was trained on DAPT2020 and that the benign rows were sampled from DAPT. A generated timestamp is not evidence of an observed campaign clock, and the release has no qualified execution or generator-realization grouping. DSRL can support a declared synthetic-development question; it cannot serve as independent confirmation of its parent simply by being a different file.

3.3.10 S-DAPT-2026

The inspected arXiv version of Tijjani et al. (2026) was withdrawn on April 1, 2026. The source audit did not acquire a corrected, qualified dataset or generator release. A later posting was noted, but corrected data access, source lineage, clock semantics, and dataset rights remained unverified. Consequently there is no measured row count, fitted result, or ROC-AUC for S-DAPT in this work. The finding is bounded to inspected sources; it is not proof that no usable artifact exists anywhere.

3.3.11 Secondary technique datasets

CasinoLimit was published by Kilian et al. (2025) at RAID; CAM-LDS has a 2026 journal version by Landauer et al. Both supply additional log-based context for the secondary T1105 experiment, but neither changes the primary UNRAVELED estimand by being listed alongside it.

CasinoLimit selector development uses 1,494 clean-calibration targets, including 87 T1105 positives, across eighteen runs. Evaluation uses 920 targets with seventeen positives across eighteen runs. CAM-LDS evaluation uses 4,209 targets with 100 positives across eighteen runs from one held-out family. The small CAM-LDS calibration set of 68 targets/eight positives is not used to fit or tune the transferred selectors. The underlying source-specific classifiers remain unchanged.

CasinoLimit targets are annotation-onset proxies; CAM-LDS targets are labeled interval-state proxies. Negative targets are other author techniques, not verified benign activity. The project had already examined both datasets. Thus this supplement provides previously exposed score-policy transfer evidence, not an untouched independent replication of exfiltration detection, a common-unit pooled accuracy, or a legitimate-background false-positive rate.

3.3.12 Source references and reproduction materials

Dataset citations and verified publication status are supplied in REFERENCES.json. Complete native counts appear in native_class_counts.csv; source qualification and exact cutoff bounds are preserved in the source snapshots. The source manifest binds the publication inputs. Raw network/host traces and private row-linked predictions are not redistributed by this publication assembly.

The completed scope is therefore explicit: one primary flow-stage measurement campaign, two secondary technique datasets with a different target, and four documented extension-qualification decisions. More files, fitting seeds, or synthetic rows do not independently broaden the number of real attack executions represented by a result.

3.4 Evaluation Anchor and Temporal Controls

The temporal experiment selects the first ceiling-half of each class within each later capture, ordered by observable-event hash. This common anchor contains 104,051 rows: 96,098 benign, 6,213 other stage, 18 movement, and 1,722 exfiltration. Both training arms and all three fitting seeds predict the same anchor.

Both eligible training pools exclude any row whose current-feature fingerprint appears in the anchor. After this purge, the earlier pool has 93,470 rows and the mixed pool 149,536. Matched fitting counts are 20,000 benign, 5,000 other-stage, 20 movement, and 1,740 exfiltration. Exact-feature purging removes one identifiable overlap channel; it does not make adjacent events independent or eliminate every source of dataset-specific dependence.

The temporal experiment uses fixed LightGBM models with 200 boosting iterations, at most 15 leaves per tree, learning rate 0.05, minimum child samples 10, L2 regularization 1, and two CPU threads. Fitting seeds are 20260923, 20260924, and 20260925. Each seed evaluates current features and current-plus-history features.

The past-only arm samples from the earlier pool. The time-mixed arm may additionally sample non-anchor later observations. Sampling is by fixed seeded hashes without replacement, with identical class counts across arms. Calibration records do not enter either pool. All earlier training flows finish before later evaluation begins.

The mixed arm deliberately has access to data unavailable to a model trained earlier. For contextual features, some later training histories can also include earlier anchor-flow observations. The treatment is therefore a change in training composition and temporal access, including those dependencies. It is not a pure causal estimate of access to future labels.

A conventional stratified random-row comparison is retained as a secondary result. It changes the test population and permits some shared current-feature fingerprints across its boundary. Its difference from the chronological arm is reported separately, rather than being called the isolated temporal effect.

3.5 Historical-Evidence and Acquisition Interventions

The history-selection experiment compares current-flow-plus-role evidence with context-assisted evidence, equal probability fusion, maximum-confidence selection, ordinary learned selection, stage-weighted selection, and context-dropout training. Four forward folds generate held-out expert predictions for selector fitting. The stage-weighted selector estimates the additional classification error associated with choosing history, using weight four for movement and exfiltration and one for the other true classes.

Five fixed evaluation conditions preserve the true labels: clean history, half missing, all missing, a five-minute-old snapshot, and wrong-host history. The last is a deliberate linkage corruption, not a measured incident. Availability and relative evidence age are observable selector inputs; the true stage is not an inference input. The selector-target weights are illustrative priorities, not a literature-derived loss ratio.

The acquisition experiment has two optional evidence groups: roles and history. Four fixed LightGBM classifiers cover current evidence and each optional subset.

Forward-held-out transition targets train greedy acquisition policies that estimate either stage-weighted error reduction or entropy reduction. Unacquired evidence values are unavailable to the policy.

Costs are one unit for roles and two for history; budgets are one, two, and three. Nominal delays are 0.25 and 0.75 with a decision deadline of one simulated time unit. Failed or late requests still consume budget. Conditions are clean delivery, delayed/unavailable evidence, and wrong-host history. Shared row-specific schedules permit paired policy comparisons. A policy ends with the classifier for the last successfully delivered subset. No unresolved case receives an automatic correct label.

These conditions probe behavior under explicit assumptions. They do not measure live collection prices, real sensor outage rates, or deployment latency. The system does not reproduce complete published acquisition architectures; those approaches establish context for the problem rather than reproduced baselines.

3.6 Implemented Models and Mathematical Summary

Implementation audit: September 24, 2026. This material describes the frozen PX080-PX083 implementations and their saved artifacts. It does not report new fitting or attribute proposed architectures to completed experiments. The accompanying machine-readable inventory binds source hashes and records configuration details, retained estimator metadata and model counts.

3.6.1 What the models do, in ordinary language

The main classifier is a collection of small decision trees. Each new set of trees improves on the scores already produced. One expert sees the current flow; another can also see summaries of earlier activity. A selector learns which expert is less likely to

make a stage mistake. A separate acquisition policy estimates whether asking for another evidence group is worth its simulated cost. The chronological experiment changes the training population while keeping the classifier configuration fixed.

The three main experiments fitted **141 LightGBM estimators: 111 multiclass classifiers and 30 regressors**. These are repeated model fits for cross-fitting, controls and seeds, not 141 different architectures. The supplementary T1105 study fitted **two Ridge regressors** and reused previously fitted binary logistic classifiers. Thus the batch contains **143 new fits**, while the T1105 models address a different target from the four-class flow experiments.

Table 3-6. Completed model-fitting inventory

Experiment	Classifier fits	Regressor fits	Derivation
PX080: choose whether to use history	33	6	Per seed: eight forward-fold experts, two final experts, one augmented expert and two selectors; three seeds
PX081: choose which evidence to request	60	24	Per seed: sixteen forward-fold subset experts, four final experts and eight transition regressors; three seeds
PX082: compare training composition	18	0	Three seeds $\times$ two feature views $\times$ three training/evaluation protocols
PX083: supplementary score-policy transfer	0 new	2	One ordinary and one target-cost Ridge selector; six existing T1105 classifier controls reused across two datasets

Counts are verified against PX080 completion, PX081 receipt, PX082 completion, and PX083 completion. The four evaluation classes in the main experiments are the prepared grouping Benign, OtherAttackStage, LateralMovement and DataExfiltration. They are **declared evaluation classes**, not a universal native taxonomy.

Movement in the inspected sensor represents author Remote System Discovery progress, not independently verified successful movement.

3.6.2 LightGBM multiclass experts

3.6.3 Simple explanation

A tree asks a sequence of questions about numeric features, such as a flow size or a count of earlier contacts. Boosting adds trees sequentially to improve the model's scores. LightGBM uses binned feature values and grows a selected leaf at a time; these are library capabilities, not algorithmic inventions in this praxis. The primary algorithm reference is Ke et al. (2017, Section 2.1); implementation details follow the versioned LightGBM developers (n.d.-a) feature documentation.

3.6.4 Mathematical summary

An additive representation of the four class scores and their softmax transformation is:

$$F_k(x) = F_{0,k} + \eta \sum_{m=1}^M h_{mk}(x),$$
$$p_k(x) = \frac{\exp(F_k(x))}{\sum_{j=0}^3 \exp(F_j(x))}.$$

Here, x is the supplied feature vector, h is a class-specific tree contribution, M is the number of boosting iterations, and η is the learning rate.

The multiclass data-fit loss and the implemented decision rule are:

$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^n \log p_{y_i}(x_i),$$

$$\hat{y}_i = \arg \max_{k \in \{0, 1, 2, 3\}} p_k(x_i).$$

Here, y is the declared class of fitting example i ; the largest class score determines its predicted class, with NumPy argmax selecting the first class in a tie.

The equation specifies the data-fit loss; LightGBM's tree-growing procedure and L2 leaf regularization determine the fitted ensemble. The code uses the ordinary multiclass objective inferred by `LGBMClassifier`, not a custom warning-retention objective. The expert fitting calls do not supply the stage cost vector as class or sample weights. Class-specific sampling caps change the fitting composition; that is different from weighting the classifier's loss. The scores are `predict_proba` outputs, but the main experiments perform no probability calibration. LightGBM developers (n.d.-b), classifier API, `objective`, `class_weight`, `reg_lambda`, `predict_proba`.

3.6.5 Exact configured settings

Table 3-7. Boosting iterations, tree size, and learning rate

Model role	Boosting iterations	Maximum leaves/tree	Learning rate
PX080 experts, including augmented comparator	180	15	0.05
PX080 selector regressors	120	7	0.05
PX081 subset experts	150	15	0.05
PX081 acquisition regressors	100	9	0.05
PX082 experts	200	15	0.05

Table 3-8. Regularization and execution settings

Model role	Minimum child samples	L2 leaf regularization	CPU threads
PX080 experts, including augmented comparator	10	1	4
PX080 selector regressors	30	5	4
PX081 subset experts	10	1	2
PX081 acquisition regressors	20	1	2
PX082 experts	10	1	2

`n_estimators` is a boosting-iteration setting. A four-class LightGBM iteration builds class-specific trees; it is inaccurate to describe 180 multiclass iterations as 180 total trees. Saved PX080 classifiers inspected in this audit have 180 iterations and 720 trees, and the retained PX081 classifiers have 150 iterations and 600 trees. PX082 saved probabilities and receipts, not serialized estimators, so its inventory records configured iterations without asserting an independently inspected final tree count. LightGBM developers (n.d.-d), parameters, `num_iterations`.

All these fits use `boosting_type='gbdt'`. Common unchanged wrapper defaults include `max_depth=-1`, `min_child_weight=0.001`, `min_split_gain=0`, `reg_alpha=0`, `subsample=1`, `subsample_freq=0`, `colsample_bytree=1`, and `subsample_for_bin=200000`. No validation-based early stopping or hyperparameter search is invoked. PX080 and PX082 explicitly set `deterministic=True` and `force_col_wise=True`; PX081 does not set those two flags. GOSS and DART were not selected. The implementations use CPU execution. PX080, PX082 and PX083 completion receipts explicitly record no AWS use, while PX081 records local CPU compute; cloud data-acquisition activity elsewhere is not

a GPU model-fitting result. Full constructor and saved-boost parameters are preserved in MODEL_INVENTORY.json.

3.6.6 PX080: learning when history adds a mistake

3.6.7 Features and honest comparison targets

The current expert sees 62 current-flow features and eight role features: 70 inputs. The context expert adds 36 history summaries, the log-transformed age of the newest available earlier event, and a history-channel availability flag: 108 inputs. The 20 selector inputs comprise the two four-class probability vectors, their four differences, each expert's maximum score/margin/entropy, and the two history-status values. These are observable prediction/status features; the true class is not supplied at inference. See implementation, functions observe, selector_features, gate_target and fit_experts (lines 59-93).

Four forward folds produce selector-training predictions: fit on captures earlier than 1 and predict capture 1, then repeat for captures 2, 3 and 4. Fitting completion times must precede validation start times. Within a fold, the same clean-fitted experts score clean, half-missing and stale histories. The ordinary and stage-cost regressors learn from these held-out comparisons; final experts are then fitted on the designated earlier period. This is forward cross-fitting, not random-fold cross-validation, nested tuning, or untouched external confirmation. The held-out comparison data belong to the already exposed development campaign.

Fitting row caps are 20,000 benign and 5,000 for each other class. Per-capture selector-validation caps are 12,000 benign and 5,000 for each other class. Seeds are

20260924, 20260925 and 20260926. Capture 5 is unused by these fits; captures 6-10 supply later descriptive evaluation. See the frozen protocol and runner lines 133-169.

3.6.8 Selector target and decision

Define weighted stage error as:

$$\begin{aligned}\ell_w(y, p) &= w_y \mathbf{1}\{\arg \max_k p_k \neq y\}, \\ w &= (1, 1, 4, 4).\end{aligned}$$

Here, w assigns a self-imposed cost to the true evaluation class; every wrong destination for that class receives the same cost.

The gate's training response and deployment choice are:

$$\begin{aligned}t_i &= \ell_w(y_i, p_i^H) - \ell_w(y_i, p_i^C), \\ p_i^{\text{gate}} &= p_i^H \quad \text{if } g(z_i) < 0 \\ p_i^{\text{gate}} &= p_i^C \quad \text{if } g(z_i) \geq 0\end{aligned}$$

Here, C and H denote the current and context experts, z contains the selector's observable inputs, and g is the fitted regression function.

The ordinary gate uses cost one for every class. A LightGBM regressor fits the numerical response using squared-error regression; it predicts a signed incremental loss, **not a calibrated probability that history is harmful**. The weight of four multiplies the response on selected true classes; it is not passed as `sample_weight` and is not a fourfold weight on the regressor's residual loss. LightGBM developers (n.d.-c), regressor API, objective; source target and fitting calls, lines 79-86 and 149-158.

This error definition treats exfiltration called benign and exfiltration called another attack stage equally: both cost four. It therefore does not explicitly optimize the

retention of an attack warning. That implementation fact helps interpret the measured tradeoff, but does not establish that a replacement loss would improve it.

3.6.9 Fixed and augmentation controls

The seven arms are current-plus-roles, context, equal probability fusion, maximum-confidence gate, ordinary learned gate, stage-harm gate and context-dropout classifier. Fusion averages the two probability vectors. The confidence gate chooses context only when its largest probability exceeds the current expert's largest probability; ties choose current. Neither control is another fitted selector.

The dropout comparator is one additional LightGBM classifier trained on two copies of each fitting row: a clean-history copy and a copy with history removed for the deterministic hash-defined half-missing subset. Each copy receives sample weight 0.5, preserving the row's total training weight. Removed history and age are zeroed and the availability flag is zero. This is feature-loss data augmentation, **not neural dropout or LightGBM DART**. Five evaluation conditions are retained: clean, half missing, all missing, a five-minute-old historical snapshot, and wrong-host history. The stale condition is a simulated cached-state intervention, not a measurement of real log-delivery delay. Source, lines 63-71, 107-113 and 162-177.

3.6.10 PX081: selecting a useful evidence request

3.6.11 What is fitted

Four multiclass experts represent the evidence already delivered: current only (62 inputs), current plus roles (70), current plus history (98), and all three groups (106). The state is a two-bit code: roles contributes bit 1 and history bit 2. Four legal transitions are

learned: request roles or history from the empty optional state, history after roles, and roles after history. Each transition has an entropy regressor and an error-reduction regressor. Their inputs are the features already present in that state plus its four class probabilities; they do not include the unacquired group's values. Dimensions are 66, 74 or 102 depending on the transition's initial state. See runner, lines 18-19, 33-59 and 172-190.

The four subset experts are cross-fitted over the same four earlier-to-later capture folds. A capped sample of the resulting held-out rows fits the transition regressors. Fitting and selector caps are 12,000 benign and 4,000 per other class. Final experts use the designated earlier fitting captures. Seeds are 8101, 8102 and 8103. The learned responses come from clean evidence; delayed/unavailable and wrong-host conditions are later simulated evaluations, not alternative labels used to train the policy.

3.6.12 Learned gains

The two responses for acquiring group a in delivered state S are:

$$u_i^{\text{harm}}(S, a) = \ell_w(y_i, p_i^S) - \ell_w(y_i, p_i^{S \cup \{a\}}),$$

$$u_i^{\text{entropy}}(S, a) = H(p_i^S) - H(p_i^{S \cup \{a\}}).$$

Here, positive u means a favorable before-to-after change. H is predictive entropy: the negative sum of each class probability multiplied by its natural logarithm. The code clips logarithm inputs at 10^{-15} .

These signs are the reverse of the history gate's incremental-loss response: acquisition seeks positive predicted improvement, whereas the history gate chooses context for negative predicted extra loss. Each transition regressor uses ordinary squared-

error fitting to predict its response. Entropy reduction measures sharper model scores; it does not certify a more accurate decision.

3.6.13 Actual greedy policy and simulated constraints

For eligible groups, the policy chooses:

$$a^* = \arg \max_{a \in \mathcal{E}(S)} \frac{q_a \hat{u}(S, a)}{c_a},$$

$$\text{request only if } \frac{q_{a^*} \hat{u}(S, a^*)}{c_{a^*}} > 0.$$

Here, c is the simulated request cost, q is a supplied availability prior, and the hatted u is the fitted gain for that transition.

The eligibility conditions are:

$$a \notin A_{\text{attempted}},$$

$$C_{\text{spent}} + c_a \leq B,$$

$$T_{\text{elapsed}} + d_a^{\text{nominal}} \leq D.$$

Here, B is the budget, D is the decision deadline, and the elapsed time accumulates realized delays after a request. The code applies a 10^{-7} numerical tolerance to deadline comparisons.

Roles cost one unit and nominally take 0.25 time units; history costs two and nominally takes 0.75. Budgets are 1, 2 and 3; the deadline is one unit. The delayed/unavailable regime supplies priors 0.8 and 0.65, respectively. Other regimes supply priors of one. Delayed requests add 0, 0.5 or 1.5 time units with simulated probabilities 0.60, 0.25 and 0.15. A request consumes cost and realized delay even if unavailable or late. The delivered state changes only if the requested evidence is available

and arrives by the deadline. There are at most two requests and no retry of an attempted group. Exact replay implementation, lines 68-112.

This is a **greedy one-step gain-per-cost heuristic**, not a globally optimized sequential planner, learned delay model, or deadline guarantee. It knows the declared regime's availability priors; it does not know the row's hidden availability/delay outcome before requesting. Multiplying by availability does not fully model the probability of timely arrival. Offline code precomputes all state scores, but the replay selects only the score indexed by the evidence actually delivered. Cost, delay and severity values are design choices rather than industry measurements or literature-mandated thresholds.

The six policies are no request, roles first, history first, random order, predicted entropy reduction and predicted stage-error reduction. All conditions and budgets are retained. An all-evidence reference is separately marked `reference_only`; it bypasses the sequential delivery constraints and is not a deployable policy. None of these acquisition arms is a reproduced SEFA or Learning-to-Measure implementation.

3.6.14 PX082: holding the model fixed while changing temporal access

This experiment uses ordinary unweighted multiclass LightGBM with current features (62 inputs) or current plus the 36 history summaries (98). It does **not** use the PX080 role, age or availability additions. Eighteen fits cover three seeds (20260923-20260925), two feature views and three protocols. No selector is fitted here. All models use argmax, with no threshold or probability calibration.

Past-only and time-mixed fitting share a 104,051-row later anchor and matched class counts of 20,000 benign, 5,000 other stage, 20 movement and 1,740 exfiltration. Anchor feature fingerprints are excluded from both fitting pools. The separate random-

row protocol reserves 55% for evaluation and changes the evaluation population. Consequently, the controlled past/mixed contrast concerns training composition and temporal access; the random-row difference cannot be interpreted as the same isolated contrast. These are evaluation controls, not a new LightGBM architecture. See protocol, design, and runner, lines 85-130.

3.6.15 Supplementary PX083: a Ridge policy on a different recognition task

3.6.16 Scope and reused experts

The target is **T1105 Ingress Tool Transfer versus other author technique labels**. The negatives are not independently verified benign flows. CasinoLimit and CAM-LDS use different annotated units. This experiment therefore does not establish movement recognition, exfiltration prediction or benign false-alarm reduction. Only the selector transfers unchanged from Casino to CAM-LDS; each dataset retains its own native experts.

The reused current, context and mixed-dropout experts are binary LogisticRegression models, not LightGBM, GNNs or language models. Their frozen settings are `C=1`, `class_weight='balanced'`, `solver='liblinear'`, `max_iter=2000`, and `random_state=20260920`. Stored models use L2 regularization (`l1_ratio=0` in their original scikit-learn 1.9.0 state). They use fixed hashed text features rather than learned language-model embeddings. Each input has 65,546 columns: two 32,768-column hashed blocks plus ten metadata columns. The original mixed-dropout comparator uses six views-clean, random 25% loss, random 50% loss, EXECVE absent, PROCTITLE absent and both command records absent-with sample weight 1/6 per view, multiplied by the classifier's automatic class-balancing weights. No such native model was refitted in

PX083. Original runner, functions `fit_lr` and `run`; original protocol, `classifier`, `text_features` and `arms`; Scikit-learn developers (n.d.-a), official logistic-regression documentation.

The logistic experts transform a linear score into a bounded binary score:

$$p(T1105 | x) = \frac{1}{1 + \exp[-(b + \beta^\top \phi(x))]}.$$

Here, ϕ is the fixed hashed-text and metadata representation, β contains learned coefficients, and b is the intercept. Fitting minimizes a class-balanced logistic loss with L2 regularization; the configured C controls inverse regularization strength. This compact score equation explains the classifier, while the saved library configuration above fixes its implementation. The scikit-learn software is described by Pedregosa et al. (2011). A bounded logistic score is not evidence of calibration on the later evaluation population.

3.6.17 Two deterministic linear selectors

Both selectors fit the same 1,494 Casino clean-calibration rows, including 87 target-positive rows. Only 11 rows have a nonzero expert-error comparison response; the full calibration sample size is not the number of informative disagreements. The ordinary response is context error minus current error at the strict score threshold 0.5. The target-cost response multiplies this difference by four for target-positive rows and one otherwise. The 12 observable inputs are the two scores, their signed/absolute difference, both binary confidences and margins, two visibility flags, threshold-decision disagreement, and the product of the scores. No scaler is fitted.

Ridge estimates the linear selector by:

$$(\hat{b}, \hat{\beta}) = \arg \min_{b, \beta} \left[\sum_{i=1}^n (t_i - b - z_i^\top \beta)^2 + 10 \|\beta\|_2^2 \right].$$

Here, t is the ordinary or target-cost error-difference response, z is the 12-feature vector, b is the fitted intercept, and the coefficient penalty is $\alpha=10$.

The solver is SVD, with `fit_intercept=True`; there is no random fitting seed or tuning search. A negative fitted linear score chooses context, otherwise current. The chosen expert's score is retained, and an alarm requires **strictly greater than 0.5**. The Ridge output is an unbounded regression score, not a probability. If neither expert observes a target, all arms force its final score to zero while retaining the row in the denominator. Original calibrated native-control thresholds remain separate descriptive controls and were not selected anew for these policies. Scikit-learn developers (n.d.-b), Ridge objective and solver documentation; PX083 source, lines 74-129.

Two fits generate evaluations of seven arms across 21 perturbation views per dataset: 294 tables in total. The 21 views comprise 15 conditions, with repeated random-loss seeds; these are not 21 datasets or three new Ridge fitting seeds. Original native models were fitted with scikit-learn 1.9.0; the new Ridge selectors use 1.7.2. PX083 consumes previously saved probabilities and does not reload those older classifiers for new inference. This implementation audit likewise used their stored attributes only; it did not recompute cross-version predictions.

3.6.18 Implemented versus proposed approaches

Table 3-9. Implemented and unexecuted approaches

Approach	Status in PX080-PX083
LightGBM four-class experts and regression selectors	Fitted in the main experiments
Fixed averaging, confidence gates and fixed acquisition order	Evaluated decision rules; no extra model fit
Missing-history and record-loss augmentation	Implemented comparator training, as specified above
Ridge score-policy transfer	Two supplementary fits on T1105
TabM, GNN/graph-machine-learning architecture, Qwen or another LLM	Not fitted or evaluated in this batch
New warning-destination loss or a guard against attack-to-benign changes	Possible future work; no performance result claimed
SEFA, Learning-to-Measure, reinforcement-learning planner	Literature context or proposed direction; not reproduced algorithms in these runs

3.7 Stage, Warning and Workload Metrics

Let $C(s,j)$ count records with true class s and predicted class j , let b denote benign, and let $N(s)$ be the number of true- s records. For an attack stage s :

- Exact-stage recall = $C(s,s) / N(s)$.
- Warning recall = $1 - C(s,b) / N(s)$.
- Missed-warning rate = $C(s,b) / N(s)$.
- Wrong-stage warning rate = warning recall minus exact-stage recall.

The three destinations - correct stage, another attack stage, and benign - partition each true attack stage. Warning recall is ordinary binary attack recall conditioned on the

true stage. The warning is a non-benign model label, not evidence that a production alert was displayed, triaged, or prevented harm.

The study also reports precision, per-class F1, unweighted macro-F1, benign false-alert counts and rates, and full confusion matrices. Available probability metrics remain in original experiment tables. ROC-AUC and average precision do not replace operating-point counts or establish a latency result.

A descriptive sign reversal occurs when a specified comparison increases macro-F1 and decreases a stage's warning recall on the same rows. Candidate-minus-baseline signs use a numerical tolerance of 1e-12 to suppress floating-point artifacts. This tolerance is not a practical-significance threshold. Every declared stage and condition is retained, including zero changes and opposite directions.

Compact reanalysis tables scale differences by 100. For recall this gives percentage points; for F1, one displayed score point means 0.01 on the original zero-to-one scale. Absolute F1 tables retain the zero-to-one scale.

For clarity, let TP, FP and FN denote true positives, false positives and false negatives for one class. Precision measures how many predictions of that class are correct; recall measures how many true members of that class were recovered. Their harmonic mean is F1. Macro-F1 averages class-specific F1 values equally, so a rare class has the same nominal weight as a common class.

$$P_s = \frac{TP_s}{TP_s + FP_s},$$

$$R_s = \frac{TP_s}{TP_s + FN_s}$$

$$F1_s = \frac{2TP_s}{2TP_s + FP_s + FN_s},$$

$$F1_{macro} = \frac{1}{K} \sum_{s=1}^K F1_s$$

With the confusion count C and benign label b defined above, the central decomposition is:

$$R_{stage,s} = \frac{C_{s,s}}{N_s},$$

$$R_{warning,s} = 1 - \frac{C_{s,b}}{N_s}$$

$$FPR_b = \frac{\sum_{j \neq b} C_{b,j}}{N_b}$$

These equations define ordinary confusion-matrix quantities. The model family does not change their meaning. The complete tables retain the original metric conventions; the paired reanalysis additionally marks unsupported true classes as undefined rather than silently excluding them.

3.8 Paired Reanalysis, Uncertainty and Sensitivity

The reanalysis compares error-focused versus entropy acquisition at every registered condition and budget, and evaluates three temporal contrasts: mixed versus past-only current features, mixed versus past-only historical features, and past-only history versus past-only current features. No model is refitted or threshold tuned. Ordered row identities, truth labels, source groups, and declared evaluation-class meanings must match before a contrast is computed.

One shared bootstrap plan per evaluation population samples its five source captures with replacement 2,000 times, using seed 20260923. All fitting seeds and contrasts on that population reuse the same group multiplicities. Confusion counts repeat

with each selected capture; equivalent whole-row replication is tested independently. The statistic remains event-weighted within a replicate, rather than becoming an unweighted average of capture scores.

Intervals are 95% percentile intervals for paired differences. If a resample contains no examples of a target stage, its recall difference is undefined; those draws are counted and excluded from that interval. Fixed-schema macro-F1 is undefined if a true class is absent from a resample. This convention is specific to the new reanalysis and does not overwrite the original temporal bootstrap convention. Every interval discloses its number of valid draws.

With five captures from one campaign, these are conditional, descriptive intervals. Captures are fragments of a shared attack workflow, not independent campaigns. Three fitting seeds quantify algorithmic variation on those same events; they do not triple the sample of attacks. No family-wide significance claim follows from selecting a favorable interval among stages, conditions, or seeds.

Bootstrap resampling is a general tool for measuring variation in a statistic (Efron, 1979). Here the resampling unit is a capture fragment, with shared multiplicities for each paired prediction comparison. That choice preserves the paired calculation but cannot make correlated fragments into independent campaigns. The reported intervals are explicitly conditional and descriptive.

The additional frozen sensitivity audit exhaustively removes each of five captures from each of 36 comparisons, holding predictions fixed, and removes each fitting seed from each of 12 three-seed means. The remaining seed metrics are averaged without pooling repeated flows. These finite omission ranges are not confidence intervals or

unseen-campaign validation. Ordered per-capture confusion signatures identify repeated sufficient statistics among acquisition comparisons; equal signatures do not prove identical row-level predictions.

3.9 Necessary Chronological-Support Test

For each native class, the start-only diagnostic identifies its second-earliest and latest recorded starts. A cutoff c with training starts strictly before c and evaluation starts at or after c can have two earlier and one later records of every class only if:

$$\max(\text{second-earliest start across classes}) < c \leq \min(\text{latest start across classes})$$

An empty interval rules out every single cutoff under that rule. It is stronger than finding one unsuccessful train/test percentage. A nonempty interval establishes necessary start-time count support only; flow completion, history availability, duplicate purging, and source-clock qualification impose additional constraints. Tied timestamps stay tied. No rows or native classes are discarded and no clock is repaired.

The support algorithm was committed before execution on the source CSVs. Its tests include tied event sequences, infeasible intervals, insufficient classes, and a counterexample distinguishing start-only from completion-time support. A separate implementation sweeps actual timestamp blocks and counts records on each side, independently checking the interval and class totals without importing the original bound function.

3.10 Reproducibility, Implementation and Compute

The three main experiments contain 141 model fits: 39 history-selection fits, 84 acquisition fits, and 18 temporal fits. A separate two-fit policy-transfer study brings the

broader project batch to 143; it is supplementary and is not counted as another movement/exfiltration experiment. Main model fitting used local CPU. Earlier cloud data-acquisition work is documented separately and did not produce an additional qualifying APT-stage replication.

The historical completion added reanalysis, source checks and document assembly without new model fits. Frozen source commits, input hashes, private row-level predictions, public aggregates, and independent audit receipts are preserved. The evidence index provides commands and access limits. The D1 \$5 future-compute ceiling is a limit, not money spent or evidence of a cloud experiment.

The September 24 GWU edition assembled existing results without additional model fitting. The September 27 edition retains that evidence and adds the 12 full-release fitting jobs described in Section 3.11. The accompanying build records bind the original input artifacts, generated tables, charts, references and final document. Human authorship review and institutional approval remain distinct from computational verification.

The fit counts above describe the historical batch. The full-release extension adds 12 fitting jobs and the AIT validation adds 42, yielding 197 recorded jobs across the distinct evidence branches. Job counts include regressors and constant-class controls and do not count independent campaigns.

3.11 Full-Release, Uncapped Extension

The original controlled history and acquisition study used eleven complete net1013x files (382,229 source rows), with class-specific fitting caps. To address the distinction between a complete prepared artifact and the complete upstream release, the new extension binds the original source inventory by SHA-256 and checks every released

flow file against its recorded byte hash, row count and right-anchored native-stage totals. The pinned author repository revision is d2ea90055d82fa448ab20588a13e3ec8bfd74816. All 173 files across 31 capture directories are included; six sensor views remain separate because simultaneous network observations are dependent. A source row is not an independent attack execution.

The extension protocol was written before its new fitting jobs. It fixes training to flows completed before June 27, 2021 UTC, calibration to flows starting on or after June 27 and completed before June 28, and testing to flows starting on or after June 28. Flows spanning a boundary are excluded and counted. Exact within-sensor observable identities combine endpoints, service ports, timestamps and numeric current-flow measurements. Identical copies with consistent labels are counted once; conflicting copies are quarantined. Near-duplicates and cross-sensor dependence remain limitations.

The parser takes stable numeric fields from the first 77 columns and native annotation fields from the final four columns; variable unquoted comma-containing application fields are not treated as additional numeric inputs. Benign remains benign; reconnaissance, foothold and cover-up map to OtherAttackStage; movement and exfiltration retain their author labels. Flow identifiers, absolute timestamps, endpoint addresses, MAC addresses, raw ports, VLAN and tunnel identifiers are excluded from predictors. Three coarse destination-service indicators are included. The roles arm adds eight coarse topology indicators from the two endpoints. An 'other' role is not an independently verified Internet or attacker designation.

Each sensor has a current-flow arm and a current-flow-plus-roles arm. Every eligible training row is used without a class cap, downsampling or class weighting.

LightGBM 4.7.0 uses 300 boosting iterations, 15 leaves, learning rate 0.05, minimum child support 10, L2 regularization 1.0, deterministic mode and seed 20260927. The extension environment uses NumPy 2.4.6, scikit-learn 1.9.0 and joblib 1.5.3; the historical environment remains recorded separately. No new history, future-event feature, source-sensor identity or alternative architecture is introduced. Prediction is multiclass argmax. A sensor with only one training class receives the corresponding constant-class probability vector: that is a degenerate control, not evidence of attack-learning ability. A probability-column alignment failure first exposed this library edge case; the implementation was corrected without changing cutoffs, source rows, features or selection thresholds.

Metrics include the complete four-by-four confusion matrix, exact-stage precision and recall, per-class F1, stage-conditioned non-benign warning recall, benign false-alert counts and rates, one-versus-rest ROC AUC and average precision where supported. Macro-F1 always averages the four declared classes, assigning zero to unsupported classes. It is not comparable across different support patterns as if they were equivalent tasks; for example, perfect benign-only prediction yields 0.25 on that fixed four-class measure. Unsupported warning recall is reported as unavailable, not as zero or success. Per-capture tables and every calibration/test prediction are retained.

A small applied review example freezes two choices on calibration alone. The score-only choice selects the higher macro-F1 arm, with ties favoring current flow. The conservative choice allows roles only if macro-F1 increases, warning recall for every supported attack stage falls by at most one percentage point, and benign false-positive rate rises by at most 0.1 percentage point. A missing calibration attack class makes the

decision provisional. These tolerances are illustrative engineering choices, not validated analyst preferences or a guarantee on later data. Both choices are evaluated on the unchanged test population without retuning.

The new source inventory, frozen protocol, per-sensor results, prediction files, model objects and independent count-based audit are retained in the accompanying evidence. One deterministic fit per arm estimates this declared contrast; it adds no campaign-level confidence interval. The original fitting-seed sensitivity remains the evidence about historical run variation.

3.12 Adapted External-Execution Validation on AIT

3.12.1 Question, source and pre-fit freeze

The added experiment tests the score, warning and false-alert comparison on a different source with entire executions held out. The acquisition objective comparison is retained, but the native taxonomy and evidence contract require an adaptation: benign, other attack and exfiltration replace the original four classes, and prior history is the only optional group. No movement label is inferred. Publisher roles, networks, ports and identities participate in source labeling and are excluded from model inputs. Thus this is an external-source execution test of an adapted configuration, not an exact PX081 replication.

All eight archives of AIT Netflow Data Set version 2, DOI 10.5281/zenodo.13168643, were downloaded and checked against publisher MD5s, local SHA256 hashes and every ZIP member's CRC (Soro et al., 2024). The laboratory generator supplies simulated legitimate traffic and repeated attack scenarios (Landauer et al., 2021, 2023). Qualification counts and timestamps were examined before fitting.

PROTOCOL.md fixes the comparisons; FREEZE.json, written before the first fit, binds code, prepared data, software, features, execution membership and seeds. No result-dependent tuning or source omission followed evaluation.

3.12.2 Complete population and execution split

The complete source contains 3,465,342 rows. Preparation excludes 942 empty/unknown labels, ten invalid numeric records and 21 exact duplicate records; no conflicting-identity group was found. All 3,464,369 eligible rows remain. Exact observable identity combines protocol, endpoints, ports, times, packet counts and byte counts. Zero identity-hash intersections occur between final training and evaluation. Probe identities are absent from the aggregated release, so near-duplicate observations of the same activity may remain within an execution.

Executions are ordered by earliest observed start, with alphabetical tie-breaking. Santos, Fox, Wardbeck, Russellmitchell, Shaw and Wheeler provide 2,397,158 training rows. Wilson and Harrison provide 1,067,211 held-out rows. The latest training completion is January 30, 2022 at 23:58 UTC, before the first held-out observation on February 3 at 00:00 UTC. Final evaluation is both execution-disjoint and calendar-separated. Development forward folds use preceding executions to predict the next one; overlapping development execution intervals mean those folds are not a strict calendar-forward stream simulation.

Table 3-10. Complete AIT source coverage and final allocation.

Counts are source observations, not independent attack counts.

Execution	Use	Raw rows	Eligible rows	Other attack	Exfiltration
santos	Train	288,547	288,536	5,937	12,848
fox	Train	351,908	351,902	56,505	14,668
wardbeck	Train	390,224	390,220	5,099	15,076
russellmitchell	Train	377,704	377,702	155,250	15,778
shaw	Train	453,541	452,598	2,357	60,898
wheeler	Train	536,206	536,200	93,393	19,177
wilson	Test	629,553	629,553	40,321	21,219
harrison	Test	437,659	437,658	76,895	23,457

3.12.3 Features, fitted models and acquisition rule

Ten current-flow features encode TCP/UDP, log duration, directional packet and byte totals, directional fractions and mean bytes per packet. Eight optional history features summarize flows from the same initiating endpoint that ended strictly before the current start and within the preceding hour: count, byte and packet totals and means, mean log duration, UDP fraction and time since the latest eligible completion. Empty histories are zero. Labels never enter history. Raw endpoint identifiers only key grouping; identifiers, ports, absolute timestamps, FQDNs, publisher roles, network names and execution identifiers are not predictors. Decisions describe completed flows.

For each of seeds 8101, 8102 and 8103, two LightGBM classifiers use current features or current plus history. Each has 150 boosting iterations, 15 leaves, learning rate .05, minimum child support ten and L2 regularization one, with deterministic execution and four CPU threads. Every eligible fitting row is used without class caps, class

weighting, tuning or early stopping. Five forward development folds supply out-of-fold probabilities. Two regressors, each with 100 boosting iterations and nine leaves, estimate entropy reduction or weighted hard-error reduction after obtaining history. The illustrative error weights are [1,1,4]. Selectors observe only current features and current probabilities; optional history and post-acquisition probabilities are excluded from their inputs. Final classifiers fit all six development executions. The inventory is 36 classifier fits and six selector fits.

Policies are current-only, always-history, acquire if predicted entropy reduction is positive, and acquire if predicted weighted-error reduction is positive. All receive the same per-row budget of two; history costs two and arrives after .75 time units before a deadline of one. These are clean-delivery offline simulations. Equal available budgets do not imply equal realized spending, so acquisition fractions are reported. The implementation uses Python 3.11.9, NumPy 2.4.6, pandas 2.3.3 and LightGBM 4.7.0; full dependency pins accompany the evidence.

3.12.4 Outcome definitions and label limits

The frozen primary direction is higher macro-F1 accompanied by lower exfiltration warning recall under error-focused versus entropy-focused selection. Every policy, seed and held-out execution is reported, along with pooled and equal-execution summaries. Three-class macro-F1 cannot be read as equivalent to the original four-class score. Warning recall counts any non-benign prediction on an exfiltration-labeled row. Exact-stage recall, complete confusion matrices, false alerts, errors and acquisition counts remain separately available. Fitting seeds do not multiply the number of independent executions; no campaign-population confidence interval is claimed.

Publisher labels use topology, service ports and attack timing. The released UDP notebook labels residual unmatched traffic as exfiltration. A descriptive audit confirms that all 183,121 native exfiltration rows use DNS port 53 and a publisher-identified attacker endpoint. This establishes consistency with the published scenario, not independent payload verification. Those fields remain excluded from predictors. The computational audit independently recomputes reported results and verifies frozen hashes, strict prior-history boundaries, fitting populations, OOF targets, acquisition actions, selected probabilities and budget constraints. It passes all 1,245 checks.

Chapter 4: Results

4.1 Primary Finding: Overall Scores and Missed Exfiltration Warnings

Table 4-1. Acquisition comparisons at the largest registered budget.

The policies share the same rows and available budget. A warning is any non-benign predicted label. False alerts use the same 192,193 benign evaluation rows; warning and exact exfiltration recall use the same 3,442 exfiltration rows. Means are across three fits.

Condition / policy	Macro-F1	Exact exfil. recall	Exfil. warning recall	Benign alerts
Clean / Entropy	0.7148	67.29%	85.18%	111.3
Clean / Error focused	0.7379	67.18%	76.25%	122.3
Delayed / Entropy	0.7180	67.43%	81.00%	144.0
Delayed / Error focused	0.7427	67.43%	69.20%	146.3
Wrong host / Entropy	0.6873	67.22%	77.52%	73.0
Wrong host / Error focused	0.6763	67.01%	77.37%	87.0

In clean replay, error-focused acquisition improves macro-F1 by 0.0231 over entropy acquisition, but warning recall for exfiltration falls by 8.93 percentage points. The number of such records dismissed as benign consequently increases. Under delayed or unavailable evidence, macro-F1 also rises while warning recall falls. The wrong-host condition and lower budgets are included in the full analysis rather than omitted when their directions differ.

This result concerns destinations of classification errors. Some entropy-policy mistakes assign exfiltration to another attack stage, preserving an attack label. Other decisions from the error-focused policy assign benign. The true-class weighted objective

charges both mistakes the same weight. The observed pattern is therefore consistent with an objective that fails to distinguish those destinations, although it does not establish that changing the loss alone would fix the behavior.

Across all registered conditions and budgets, 19 of 27 paired seed comparisons raise macro-F1 while reducing exfiltration warning recall. In 17 of 27, the descriptive macro-F1 interval lies above zero and the warning-recall interval below zero. These comparisons share data and fitted components; budget-one clean and wrong-history outcomes are identical because that budget cannot acquire history. They are correlated repetitions of specified comparisons, not 27 independent tests or campaigns.

Table 4-2. Acquisition differences by condition and budget

Condition	Budget	Mean Δ F1 (x100)	Mean Δ exfil warning (pp)	Mean extra lost warnings	F1 up / warning down seeds
clean	1	3.61	-20.12	692.67	2
clean	2	3.70	-20.26	697.33	2
clean	3	2.31	-8.93	307.33	2
delayed unavailable	1	3.42	-13.81	475.33	2
delayed unavailable	2	3.57	-13.95	480.00	2
delayed unavailable	3	2.47	-11.80	406.00	3
wrong host history	1	3.61	-20.12	692.67	2
wrong host history	2	12.34	-20.33	699.67	3
wrong host history	3	-1.10	-0.15	5.33	1

Clean budget-three means conceal marked fitting-seed variation. Exfiltration-to-benign counts range from 203 to 1,105 for entropy and 206 to 1,127 for harm. The paired increases are 897, 3 and 22 lost warnings, respectively (mean 307.33). Macro-F1 improves in two seeds and declines slightly in one. For seed 8101, exact exfiltration errors change only from 1,122 to 1,130, while wrong-attack-stage predictions fall from 900 to 11 and benign predictions rise from 222 to 1,119. This concrete error-destination shift explains why nearly unchanged exact exfiltration recall can coexist with a large loss of warnings. It does not identify a generally effective policy.

Table 4-3. Clean budget-three differences by fitting seed

Seed	Δ F1 (x100)	Exfil $\rightarrow$ benign	All exact exfil errors	Δ warning pp [95% paired interval]
8101	5.65	222 $\rightarrow$ 1119	1122 $\rightarrow$ 1130	-26.06 [-92.42, -0.88]
8102	-0.08	203 $\rightarrow$ 206	1128 $\rightarrow$ 1129	-0.09 [-0.52, 0.00]
8103	1.38	1105 $\rightarrow$ 1127	1128 $\rightarrow$ 1130	-0.64 [-3.06, -0.15]

All three attack stages, including zero changes and reverse directions, are included in the complete paired table. The full set was specified before this reanalysis, after selected warning-loss observations had already been inspected; it remains retrospective.

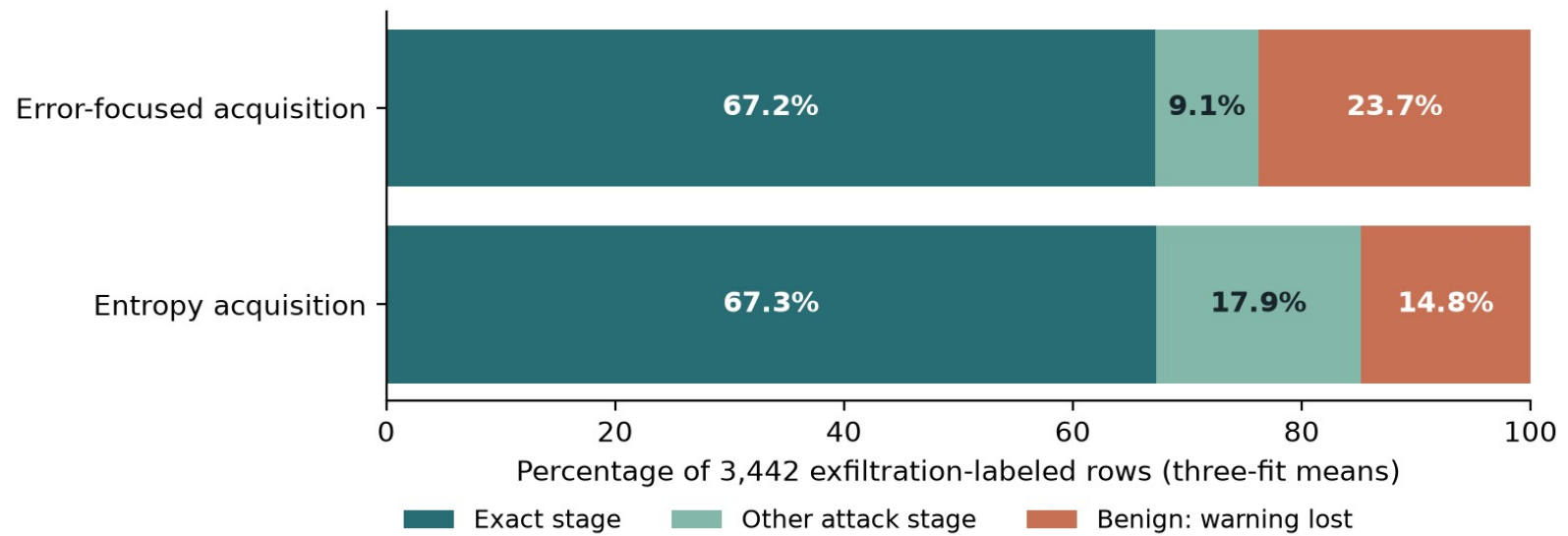
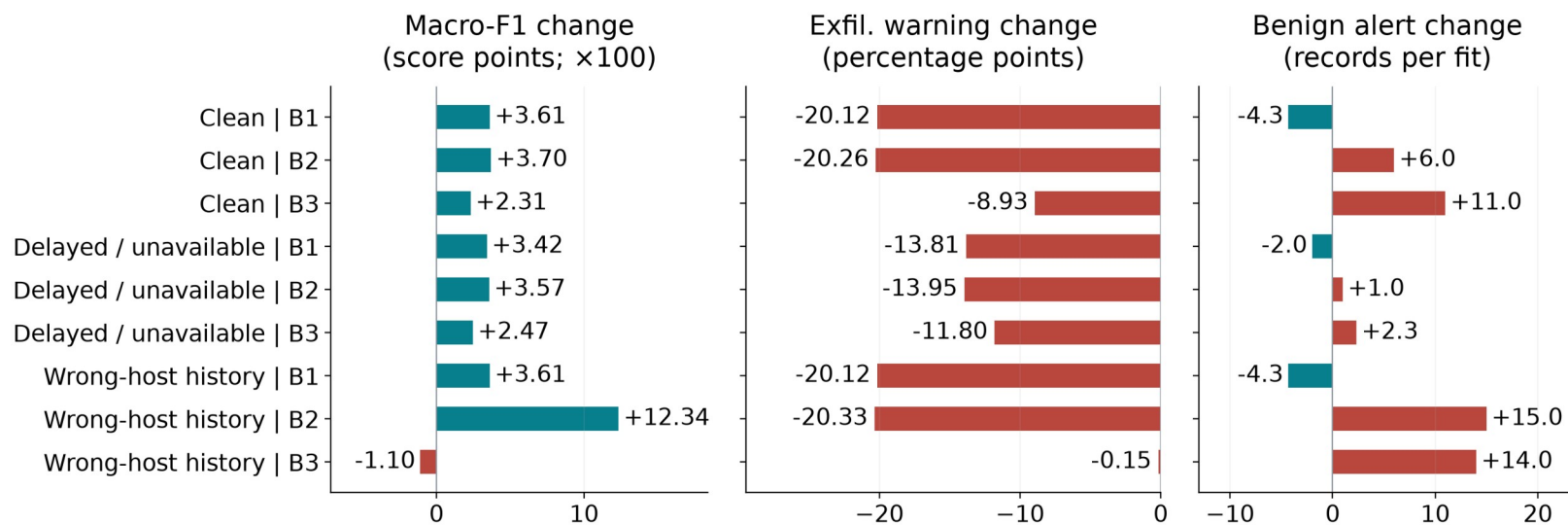


Figure 4-1. Exfiltration error destinations in clean replay at budget three. Exact-stage predictions and wrong-stage attack predictions both retain a warning; benign predictions do not. These are model-label outcomes on the same author-labeled records.

Higher overall scores can accompany fewer exfiltration warnings



Error-focused minus entropy acquisition; means of 3 fits on 208,094 shared flows.
 Exfiltration denominator: 3,442. Benign denominator: 192,193.
 Costs, delays, and failures are simulated.

Figure 4-2. Complete acquisition comparisons. Each condition and budget is shown with paired F1, exfiltration warning and benign false-alert changes; all fitting seeds remain in the accompanying tables.

Across the complete acquisition inventory, the error-focused policy has higher mean macro-F1 in eight of nine condition/budget groups and lower exfiltration warning recall in all nine. Wrong-host budget three has both lower F1 and lower warning recall. The complete tables retain that unfavorable combination, the unrestricted references and every declared policy. Lower simulated request expenditure is reported as a replay result, not a measured collection saving.

4.2 Chronological History Gains and Their Tradeoff

Under past-only fitting, adding history improves macro-F1 from 0.7365 to 0.7582, movement F1 from 0.2091 to 0.2836, and exfiltration F1 from 0.7565 to 0.7671. Mean benign false alerts decrease from 24.0 to 14.3. These are positive changes under the stricter training protocol.

The exfiltration warning outcome moves differently: exfiltration-to-benign mistakes increase in each fitting seed. Thus, even a comparison with both better F1 and fewer benign alerts can lose some warnings for a specific attack stage. The size and uncertainty of that loss belong beside the favorable metrics, rather than being removed from a positive account.

On the same 1,722 exfiltration-labeled anchor rows, adding history increases benign predictions by 7, 6 and 8 across the three fitting seeds. Mean warning recall decreases by 0.41 percentage points, while macro-F1 rises by 2.17 points. This is a much smaller warning loss than the acquisition contrast. Only the final seed's warning-difference interval excludes zero; the other two include zero. Both favorable and unfavorable outcomes should be read at that scale.

Table 4-4. Chronological history differences by fitting seed

Fitting seed	Δ macro-F1 x100 [95% interval]	Exfil $\rightarrow$ benign, current $\rightarrow$ history	Δ exfil warning pp [95% interval]
20260923	2.98 [0.42, 6.41]	565 $\rightarrow$ 572	-0.41 [-2.62, 0.00]
20260924	1.08 [-2.11, 4.20]	563 $\rightarrow$ 569	-0.35 [-2.09, 0.00]
20260925	2.45 [0.43, 3.07]	563 $\rightarrow$ 571	-0.46 [-2.88, -0.09]

Intervals for the added missed-warning counts are [0, 17], [0, 14] and [1, 17], respectively. They reuse the same capture draws as the temporal contrasts. These are descriptive fragment-resampling intervals, not independent-campaign or prospective validation. The reanalysis audit verifies unchanged input hashes, paired identities and independently recomputed arithmetic.

4.3 Supporting Finding: Fixed-Anchor Temporal Sensitivity

Table 4-5. Temporal experiment.

Values are means of three fits. Primary past/mixed rows share the 104,051-row anchor with 96,098 benign rows; random-row evaluations contain 210,226 rows with 191,515 benign rows. Benign alert rates accompany counts because these denominators differ.

Movement denotes the author label described in Chapter 3.

Features / fit	Macro-F1	Movement recall	Movement F1	Exfil. F1	Benign alerts (rate)
Current / Past	0.7365	25.93%	0.2091	0.7565	24.0 (0.025%)
Current / Mixed	0.7997	61.11%	0.2582	0.9523	88.7 (0.092%)
Current / Random	0.8093	57.84%	0.3116	0.9404	84.0 (0.044%)
Current + history / Past	0.7582	29.63%	0.2836	0.7671	14.3 (0.015%)
Current + history / Mixed	0.7964	57.41%	0.2798	0.9092	47.7 (0.050%)
Current + history / Random	0.8507	64.71%	0.4336	0.9753	49.3 (0.026%)

With current features, time-mixed fitting increases macro-F1 by 0.0632 and exact movement-label recall by 35.19 percentage points. Adding history yields a mixed-minus-past macro-F1 increase of 0.0383. These effects arise while the classifier family, anchor rows, and class fitting counts are held fixed. They establish sensitivity to the declared training-pool change on these observations.

The higher scores also accompany more benign false alerts: the current-feature mean count increases from 24.0 to 88.7. A model ranking based on F1 alone would conceal this workload change. The conventional random-row score is retained but is not used to estimate the controlled temporal effect.

The complete retrospective reanalysis gives positive current-feature macro-F1 differences in all three seeds, with each descriptive capture interval above zero. For current-plus-history, all point differences are positive, but the final seed interval spans zero. The intervals vary widely because the resampling units are only five capture fragments from the same campaign; they do not establish a population-wide inflation factor.

Table 4-6. Fixed-anchor temporal differences by fitting seed

Features	Fitting seed	Δ macro-F1 (x100)	95% F1 interval (x100)	Δ exfil warning (pp)
Current	20260923	7.52	[1.96, 26.31]	32.11
Current + history	20260923	3.32	[1.20, 17.24]	12.66
Current	20260924	6.99	[1.65, 24.36]	32.00
Current + history	20260924	2.97	[0.62, 15.49]	9.29
Current	20260925	4.46	[0.12, 20.62]	31.94
Current + history	20260925	5.18	[-0.75, 25.69]	31.59

For every comparison, 1,981 of 2,000 draws support fixed-four-class macro-F1 and movement recall; 19 draws omit the true movement class and are excluded from those intervals. All 2,000 draws support exfiltration warning recall and benign false-alert rate. The same capture multiplicities are reused across all seeds and contrasts. These new support-conditional intervals are distinct from the original temporal experiment's 1,000-draw bootstrap. Every stage, count and paired interval is retained in the complete reanalysis.

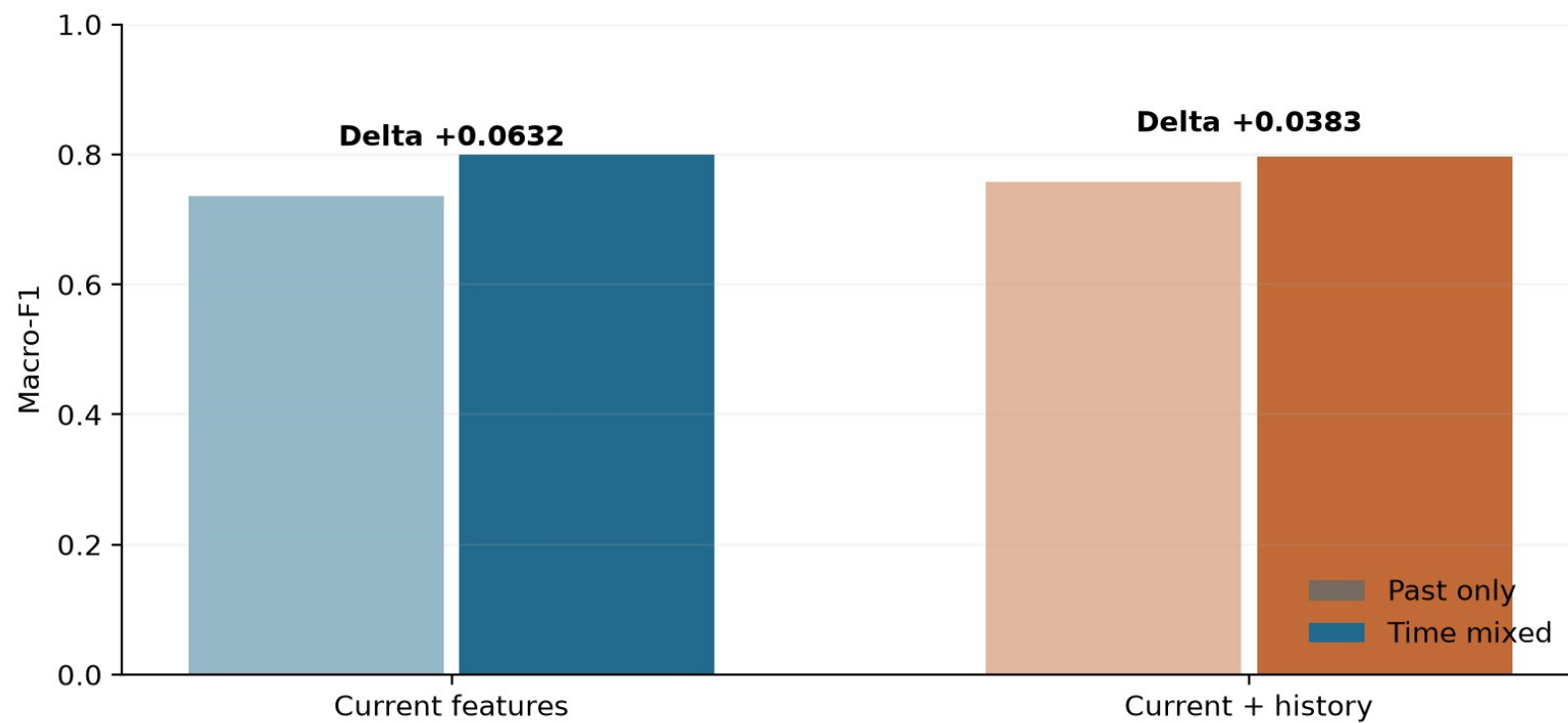


Figure 4-3. Fixed-anchor training-composition differences. Bars show means of three fits on the same events. The adjacent table reports per-seed conditional capture-bootstrap intervals.

4.4 History Selection and Simple Controls

The stage-weighted history selector improves movement-label recall over ordinary selection in all five conditions, by 4.76 to 14.29 percentage points. It also increases benign false alerts and weighted classification error in every condition. The current-plus-roles classifier retains higher movement recall than the weighted selector throughout. It is a learned LightGBM baseline, not a deterministic rule.

Table 4-7. History-selection tradeoffs.

Recall and false alerts are three-fit means. The same 35 movement rows and 192,193 benign rows recur across conditions.

Condition	Ordinary recall	Weighted recall	Dropout recall	Alerts: ordinary / weighted
Clean	64.76%	69.52%	51.43%	85.7 / 109.0
Half missing	52.38%	60.95%	62.86%	74.3 / 93.7
All missing	36.19%	42.86%	68.57%	62.3 / 77.0
Stale	49.52%	63.81%	48.57%	85.0 / 113.3
Wrong host	35.24%	44.76%	68.57%	61.7 / 81.3

With all history missing, the dropout-trained comparator reaches 68.57% movement recall versus 42.86% for weighted selection. The acquisition policy lowers simulated spending by 32.24% relative to entropy acquisition in clean, largest-budget replay, but does not consistently reduce the weighted error outcome. These results show why a measurement praxis should retain simple controls and multiple outcomes. They do not require a claim that complex detectors generally fail.

Context selection: movement recall and false-alert tradeoffs

	Movement recall (%) Higher is better; denominator 35					Benign false alerts Lower is better; denominator 192,193				
Current + roles	78.1	78.1	78.1	78.1	78.1	139.3	139.3	139.3	139.3	139.3
Context	52.4	29.5	0.0	21.9	0.0	48.3	28.7	0.0	37.7	0.0
Equal fusion	60.0	44.8	18.1	41.9	19.0	76.0	54.3	29.7	67.7	32.3
Confidence gate	60.0	44.8	18.1	41.9	19.0	77.0	54.3	29.7	69.0	32.7
Ordinary gate	64.8	52.4	36.2	49.5	35.2	85.7	74.3	62.3	85.0	61.7
Stage-harm gate	69.5	61.0	42.9	63.8	44.8	109.0	93.7	77.0	113.3	81.3
Context dropout	51.4	62.9	68.6	48.6	68.6	68.7	95.0	116.0	77.7	114.7
	Clean	Half missing	All missing	Stale 5 min	Wrong host	Clean	Half missing	All missing	Stale 5 min	Wrong host

Every cell is a mean of 3 fits on the same flows; darker shading means more of that quantity.
 Movement labels denote Remote System Discovery on one host pair.
 Missing, stale, and wrong-host histories are simulated.

Figure 4-4. History-selection movement recall and benign false-alert outcomes across all five evidence conditions. Stage weights affect selector targets, while the current-plus-roles model supplies a strong simple control.

4.5 Dataset Evaluation and Qualification Results

Table 4-8. Results of source qualification.

These counts describe the inspected files; eligibility refers to the unchanged all-native-class temporal comparison.

Source	Inspected rows	Native classes	Completed finding
SCVIC-APT-2021	259,120	6	Possible recorded-start support; clocks and execution mapping unqualified
DAPT2020	86,691	5	No all-class single cutoff; 15 exfiltration rows
DSRL-APT-2023	65,000	5	DAPT-derived synthetic source; no natural event chronology
S-DAPT-2026	None acquired	Not verified	No qualified acquired release

For DAPT2020, the cutoff would need to be later than July 19, 2019, 16:38:37 to include two earlier exfiltration examples, yet no later than July 17, 2019, 19:24:55 to retain a reconnaissance example in evaluation. The conditions contradict each other. The diagnosis uses all 86,691 rows and the five native classes; it is not caused by choosing an inconvenient 60/15/10/15 split. The release can support other questions, including a separately designed unknown-stage evaluation, but the unchanged closed-set temporal comparison cannot be run honestly on those bytes.

SCVIC's recorded-start necessary interval is nonempty: after October 21, 2015, 10:21:12 through 22:56:16. The support test therefore does not reject every possible SCVIC cutoff. However, 220 benign rows have 1970 dates, remaining rows have 2015 dates, time resolution is mixed, and a source-supported row-to-execution map and physical-clock interpretation were not obtained. The author-listed test file is absent from

the inspected local sources. The completed analysis is a recorded-time support diagnostic, not a certified temporal model evaluation.

DSRL's 65,000-row artifact combines synthetic attacks derived from DAPT with 10,000 benign rows sampled from DAPT. Its generated timestamps do not demonstrate real event order, and its shared source prevents treating it as independent real-campaign confirmation. S-DAPT contributes an availability result: the inspected arXiv record remains withdrawn and no verified corrected, accessible data artifact was acquired. A separate inaccessible listing does not resolve that status.

A separate CSV-parser implementation enumerated all tied-time membership states: 0 of 28,941 DAPT states met the all-native-class two-earlier/one-later rule, whereas 9,475 of 11,693 SCVIC states met recorded-start support. The latter did not resolve its clock or execution-identity limitations. Native-class totals and input hashes matched before and after the separate reads. The verifier did not import the original bound function. Its verification receipt records all 16 checks and source identities.

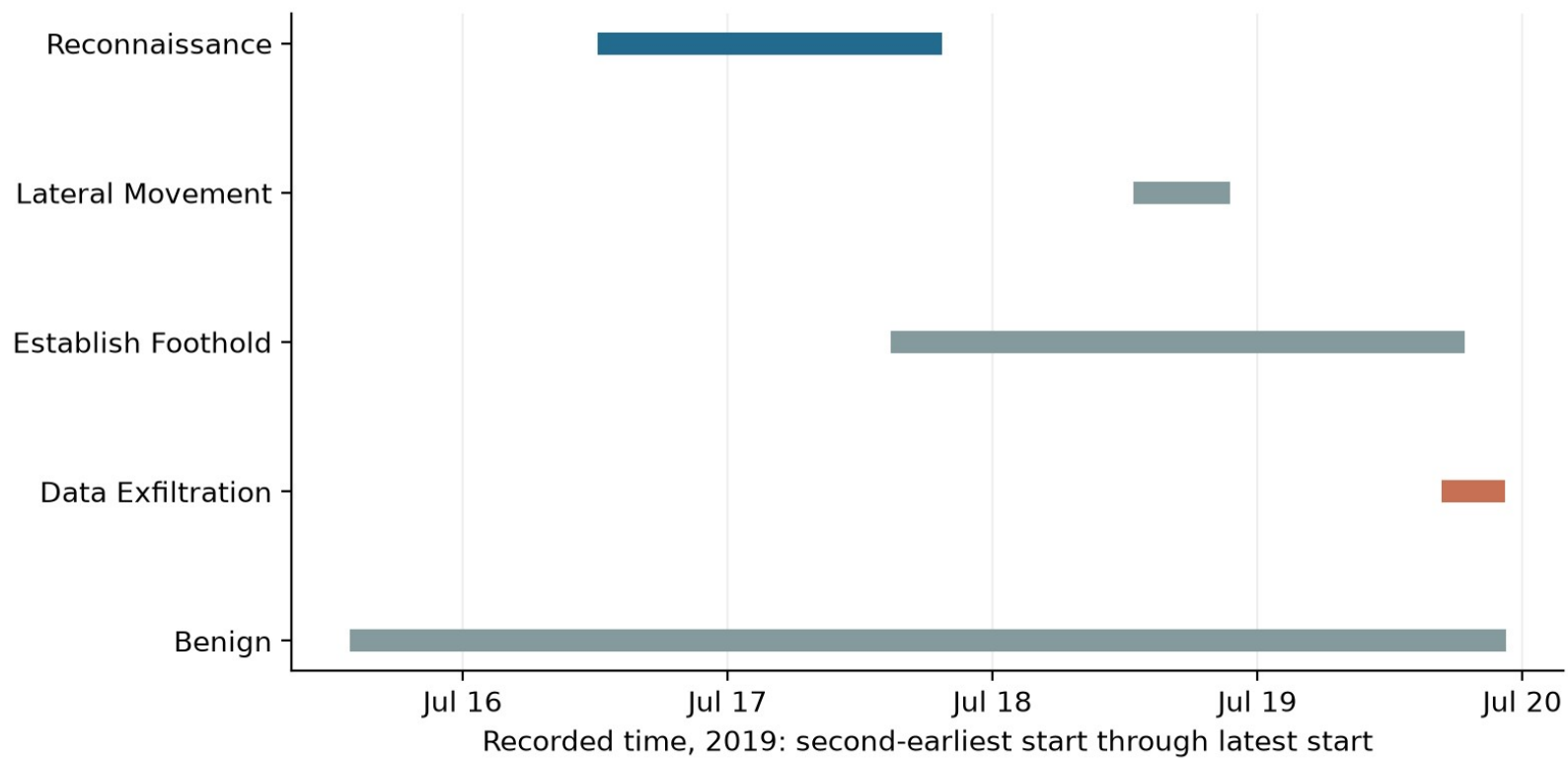
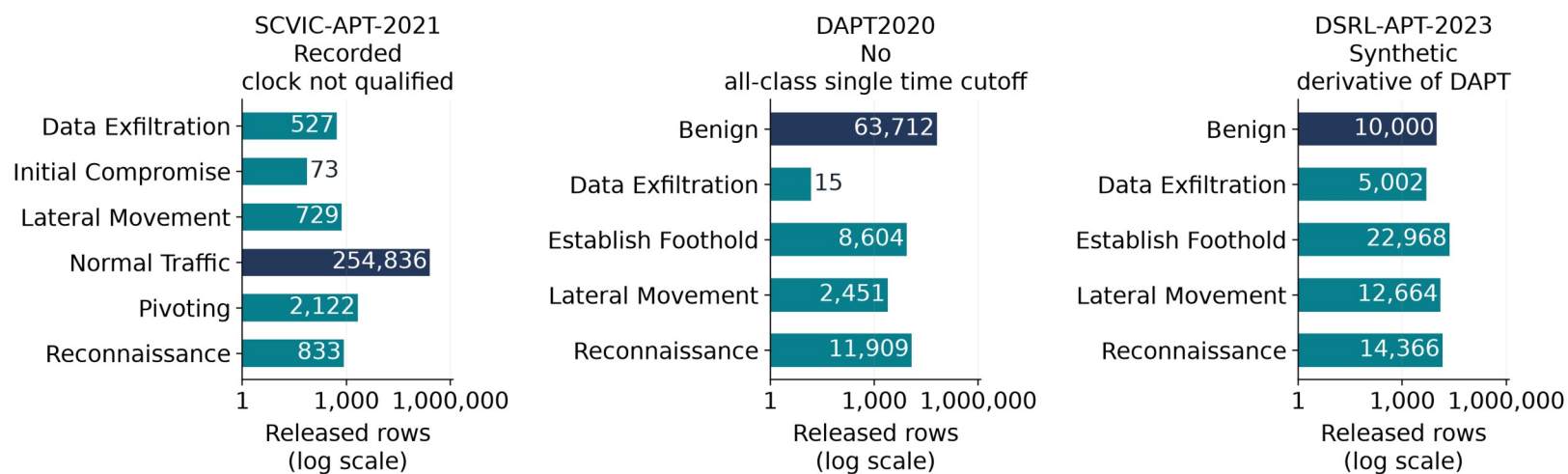


Figure 4-5. Why DAPT cannot support the unchanged all-stage cutoff. Ranges run from each native class's second-earliest to latest recorded start. A cut that trains on exfiltration cannot retain later reconnaissance in this release.

Row counts do not establish independent temporal evaluation



Native labels and counts from inspected release bytes; DSRL derives from DAPT.
S-DAPT has no bar because no qualified dataset bytes were acquired.
An unavailable measurement is not zero rows.

Figure 4-6. Observed class support and source eligibility. Dataset counts and qualification decisions describe the inspected artifacts and the declared task.

4.6 Capture and Fitting-Seed Sensitivity

The frozen sensitivity audit (e9de91d) includes every registered pair: 180 leave-one-capture-out calculations and 36 leave-one-fitting-seed-out means. Predictions are fixed, and the remaining seed metrics are averaged without pooling flows. Omission ranges are finite sensitivity summaries, not confidence intervals or independent replications.

For clean budget three, higher mean F1 and lower exfiltration warning recall remain in all five capture omissions and all three fitting-seed omissions. The magnitude is strongly sensitive to seed 8101: excluding it reduces the warning loss from 8.93 to 0.36 percentage points and the mean extra missed warnings from 307.33 to 12.50. Thus the qualitative mean tradeoff survives this deletion, but the original mean is not a stable estimate of its magnitude.

Macro-F1 changes in the tables are score points (raw score differences multiplied by 100); warning-recall changes are percentage points.

Table 4-9. Seed and capture omissions at clean budget three

Omitted unit	Δ macro-F1 ($\times 100$)	Δ exfil warning pp	Mean extra missed warnings
None (three-seed mean)	2.31	-8.93	307.33
Fit seed 8101	0.65	-0.36	12.50
Fit seed 8102	3.51	-13.35	459.50
Fit seed 8103	2.78	-13.07	450.00
Capture 6 (three-seed mean)	1.57	-25.71	305.67
Capture 7 (three-seed mean)	3.67	-8.95	307.33

Omitted unit	Δ macro-F1 ($\times 100$)	Δ exfil warning pp	Mean extra missed warnings
Capture 8 (three-seed mean)	2.26	-9.12	293.00
Capture 9 (three-seed mean)	2.16	-7.74	246.33
Capture 10 (three-seed mean)	1.52	-2.80	77.00

At individual-seed level, the clean budget-three tradeoff occurs in 10/15 capture-omission pairs: all five for seed 8101, none for seed 8102, and all five for seed 8103. Across all 12 group means, the nine original F1-up/warning-down groups retain that direction under every specified single-seed and single-capture omission; this stability statement concerns means on the same campaign.

Current-feature mixed-minus-past F1 stays positive in all 15 fitting-seed/capture-omission pairs. Per-seed ranges are:

Table 4-10. Current-feature temporal gains after capture omission

Fitting seed	Positive omission differences	Δ macro-F1 ($\times 100$) range
20260923	5/5	4.80 to 21.78
20260924	5/5	4.29 to 20.56
20260925	5/5	1.83 to 15.95

The history-feature mixed-minus-past comparison is also positive in 15/15 pairs. Chronological history-minus-current is positive in 14/15; seed 20260924 becomes -0.13 macro-F1 score points when capture 9 is omitted. No omission loses support for a declared evaluation class (unsupported omission IDs: none).

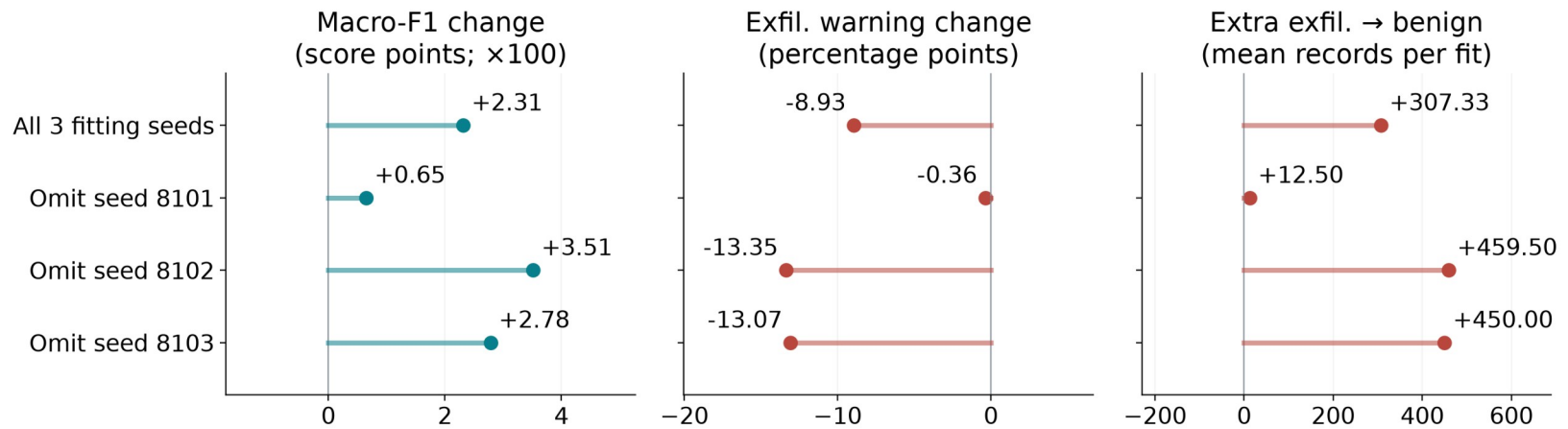
The 27 PX081 contrasts form 24 distinct ordered per-capture confusion signatures. The higher-F1/lower-warning count is 19/27 registered pairs or 17/24

aggregate-equivalent signature classes. These are not independent replication counts.

Equal confusion signatures do not establish identical individual predictions; all signatures still share events, models and one previously examined campaign.

All omitted-capture identities, per-seed supports, directions and numerical values remain in the complete sensitivity report and arithmetic receipt. Original main-study results and their limitations are unchanged.

One fitting seed accounts for most of the headline warning loss



Clean budget 3; error-focused minus entropy acquisition. Fixed predictions, not refits.

Without seed 8101: +0.65 F1 score points; -0.36 warning percentage points.

Omission ranges are not confidence intervals.

Figure 4-7. Fitting-seed influence on the clean budget-three warning tradeoff. The full mean and each leave-one-seed-out mean expose the effect-size dependence.

4.7 Supplementary Technique-Policy Transfer

The secondary study targets **T1105, Ingress Tool Transfer**. It is not a replication of lateral-movement recognition or exfiltration warnings. Negative examples have other author technique labels, so their flags cannot be called benign false alarms. CasinoLimit onset proxies and CAM-LDS labeled interval proxies also represent different target units; results are not pooled across them.

Secondary T1105 transfer: benefits and costs vary by condition

	CasinoLimit 17 T1105 / 903 other-label targets		CAM-LDS 100 T1105 / 4,109 other-label targets	
clean	+0.00	+1.00	+1.00	-5.38
random 25	-3.92	-2.14	+12.33	+10.98
random 50	-7.84	-4.61	+23.00	+22.15
random 75	-3.92	-6.61	+17.33	+20.09
support burst 60	+5.88	+1.44	+8.00	+8.66
command records absent	-17.65	-6.64	+15.00	+16.16
delay 30 deadline 0	-17.65	-6.87	+4.00	+3.53
delay 30 deadline 30	+0.00	+1.00	+1.00	-5.38
delay 120 deadline 0	-17.65	-6.64	+15.00	+16.16
delay 120 deadline 30	-17.65	-6.64	+8.00	+9.10
delay 120 deadline 120	+0.00	+1.00	+1.00	-5.38
execve absent	+0.00	-1.33	-10.00	-15.09
proctitle absent	+0.00	+0.33	+21.00	+19.40
syscall absent	-5.88	-4.54	+44.00	+46.09
path absent	+5.88	+1.55	-2.00	-4.06
	T1105 recall Δ (higher is better)	Other-label flag Δ (lower is better)	T1105 recall Δ (higher is better)	Other-label flag Δ (lower is better)

Target-cost minus current expert, percentage points. Red means an increase, not uniform benefit.

Random loss averages 3 perturbations; other conditions are deterministic.

Other-label flags are not benign false alarms.

Figure 4-8. Secondary policy transfer across all conditions

Figure. Secondary transfer under all fifteen condition names. Values are target-cost gate minus current expert in percentage points. A positive recall difference and a negative other-label flag difference are favorable in their respective columns; colors indicate the numerical sign, not a common benefit direction. Random-loss conditions average three perturbations of the same events. Other conditions use one deterministic view.

On clean CasinoLimit data, the current expert and ordinary gate both achieve F1 0.7368, recall 82.35%, and seven other-label flags. The target-cost gate and context expert achieve F1 0.5957 with the same recall and sixteen flags. On clean CAM-LDS, the current expert achieves F1 0.0495, recall 87%, and 3,329 other-label flags; the target-cost gate/context expert achieves F1 0.0534, recall 88%, and 3,108 flags. High target recall therefore coexists with poor discrimination against other technique labels.

A saved post-result diagnostic found only eleven nonzero expert-error comparisons among 1,494 CasinoLimit calibration rows. The target-cost selector's hard decisions match the context expert in every one of the 21 views on each source; different probability vectors do not establish a different operational decision. The ordinary selector matches the current expert in all CasinoLimit views and nine CAM-LDS views. These findings do not support a new adaptive advantage. Original native thresholds are preserved separately; they were not used to tune the transferred policies.

4.8 Ranking Metrics and Operating Decisions

The supplements retain per-class average precision and ROC-AUC wherever the original source reported them. They are not reconstructed from hard labels or replaced with approximations. The normalized CSV uses blank fields when a metric was

unavailable; source JSON preserves original null values. A class can have strong ranking performance while precision or its selected operating decision remains problematic.

For a concrete existing result, the clean current-plus-roles fit with seed 20260924 identifies 26 of 35 movement-labeled flows. Its movement ROC-AUC is 0.998918, average precision is 0.210308, precision is 0.184397, and recall is 0.742857. The 141 movement predictions include 26 correct labels. This is an actual aggregate from the saved evaluation, not a fabricated example or a production alert rate. Reporting ROC-AUC alone would not describe that operating-point burden. No new ROC curves are inferred from the aggregate tables.

Table 4-11. Stage-level discrimination under chronological fitting.

Each entry is the arithmetic mean of three fitting-seed metrics on the same 104,051-row anchor. Current means current-flow features; History adds earlier-activity summaries.

These class metrics are not pooled across seeds. Movement retains the narrow author-label interpretation.

Feature view / class	Precision	Exact recall	F1	ROC-AUC	Avg. precision
Current / Benign	0.9938	0.9998	0.9968	0.9988	0.9999
Current / Other attack	0.9997	0.9679	0.9836	0.9967	0.9886
Current / Movement	0.1788	0.2593	0.2091	0.9992	0.2035
Current / Exfiltration	0.8644	0.6725	0.7565	0.9680	0.6688
History / Benign	0.9934	0.9999	0.9966	0.9959	0.9996
History / Other attack	0.9986	0.9726	0.9855	0.9999	0.9991
History / Movement	0.2758	0.2963	0.2836	0.9910	0.1474
History / Exfiltration	0.9081	0.6640	0.7671	0.9780	0.7500

ROC-AUC measures ranking across score thresholds: how often a true member of one class is ranked above a nonmember. Average precision summarizes precision across recall levels and is useful when a class is rare; it is not interchangeable with trapezoidal area under a precision-recall curve. Neither ranking summary fixes the argmax decision or the resulting warning destination. The reported areas are retained source metrics; no curve is reconstructed from a single confusion matrix.

4.9 Answers to the Research Questions

RQ1: The completed comparisons demonstrate that an improved overall score can accompany fewer exfiltration warnings on the same records. In the clean budget-three acquisition comparison, mean macro-F1 rose while warning recall fell and benign false alerts increased. Chronological history also produced useful gains and a smaller warning tradeoff. Error destinations and benign workload are therefore material to interpretation.

RQ2: The fixed-anchor comparison demonstrates sensitivity to training composition and temporal access. The current-feature macro-F1 difference was +0.0632, and movement-label recall increased by 35.19 percentage points. Architecture, evaluation rows and class counts were controlled; every property of the training distributions was not isolated.

RQ3: The inspected DAPT artifact cannot satisfy the unchanged all-native-class single-cut support rule. SCVIC has possible recorded-start support but unresolved physical chronology. DSRL derives from DAPT, and no qualified S-DAPT release was acquired. These are source-specific qualification results, not four fitted replications.

The numerical comparisons answer the declared questions within the inspected records. Neither a minimum favorable score nor a self-imposed 90% recall threshold determines whether a finding is retained.

4.10 Full-Release Results and Calibration Review

The source audit accounts for all **6,877,157 rows in 173 files**. Across the six sensor views, 0 exact duplicate rows were removed, 0 conflicting-label rows were quarantined and 34 retained rows crossed a temporal boundary. The resulting analysis

contains 5,353,688 training rows, 39,399 calibration rows and 1,484,036 test rows. These are sensor observations rather than independent-event totals. All eligible training rows were used in each arm. The independent computational audit passed 622 checks of source coverage, row conservation, temporal separation, unique identities, prediction populations, probability normalization, confusion matrices, macro-F1, class support and warning counts.

Table 4-12. Complete release coverage and chronological allocation.

Source rows precede duplicate and boundary handling. Training, calibration and test counts are after handling. The three numbers in the final column follow that order.

Sensor	Files	Source rows	Train / cal / test
net1011x	28	1,792,756	1,466,865 / 5,527 / 320,359
net1012x	29	734,331	597,531 / 5,132 / 131,664
net1013x	30	1,072,368	851,355 / 8,683 / 212,323
net1014x	30	210,371	163,270 / 2,966 / 44,135
net1015x	30	175,662	144,218 / 2,790 / 28,646
netgw	26	2,891,669	2,130,449 / 14,301 / 746,909

The single-class training sensors are net1011x, net1012x, net1014x. Their constant-class outcomes document a support limit. Sensors with all four mapped classes represented in both training and testing are net1013x. Class support is detailed in Appendix D; more files do not automatically supply every attack stage at every cutoff.

Table 4-13. Full-release paired test results.

Current and roles use identical evaluation rows within a sensor. Macro-F1 uses a fixed four-class denominator. Unsupported exfiltration recall is not estimable. False-alert counts must be read with the benign support in Appendix D.

Sensor	Macro-F1 current / roles	Exfil warnings current / roles	Benign alerts current / roles
net1011x	0.2500 / 0.2500	Unsupported / Unsupported	0 / 0
net1012x	0.2500 / 0.2500	Unsupported / Unsupported	0 / 0
net1013x	0.7574 / 0.7598	71.03% / 71.06%	13 / 11
net1014x	0.2414 / 0.2414	Unsupported / Unsupported	0 / 0
net1015x	0.3268 / 0.3250	0.00% / 0.00%	12 / 13
netgw	0.4991 / 0.4991	Unsupported / Unsupported	2105 / 2104

Roles improve test macro-F1 in 2 of the six views. Views with a macro-F1 increase and a decline in warning recall for at least one supported attack stage are netgw. At netgw, macro-F1 changes from 0.49909446 to 0.49911102, benign false alerts decrease from 2,105 to 2,104, and OtherAttackStage warnings decrease by one: missed warnings increase from 25,429 to 25,430 among 26,712 such rows. This tiny one-event reversal is not evidence of a stable or practically large effect. At net1013x, both exfiltration warning recall and macro-F1 improve slightly; that arm does not reproduce the historical exfiltration-warning loss. This is the declared role-feature contrast under uncapped chronological fitting. It neither estimates the historical acquisition-budget effect on the expanded release nor converts sensor views into independent replications. The historical warning-loss finding retains its own population and sensitivity limits.

Table 4-14. Calibration-only model choices.

A review with absent attack-stage support remains provisional. Test results for the selected arms are exactly the corresponding Table 4-13 arm outcomes; no second test optimization occurs.

Sensor	Score choice	Review choice	Missing calibration stages
net1011x	current	current	Other, Movement, Exfiltration
net1012x	current	current	Other, Movement, Exfiltration
net1013x	current	current	Movement
net1014x	current	current	Movement, Exfiltration
net1015x	current	current	Other, Movement, Exfiltration
netgw	current	current	Exfiltration

This applied example demonstrates an executable audit decision and reveals which stages its calibration data can assess. 6 of the six views lack at least one attack stage in calibration. Their review decisions are therefore provisional; the extension does not demonstrate a fully supported all-stage calibration safeguard. A favorable calibration decision does not establish later warning preservation, an optimal tolerance, or actual analyst benefit. The complete per-capture confusion matrices and stage metrics accompany the table, including the unfavorable outcomes.

4.11 Adapted AIT Validation Results

The frozen primary direction did not recur: error-focused acquisition had higher macro-F1 with lower exfiltration warning recall in zero of six execution-by-seed comparisons and zero of three pooled seed comparisons. This is an adverse generalization

result for the original named-policy contrast under the adapted task. It is retained without modifying the learned policies or decision thresholds.

Table 4-15. Pooled AIT held-out results.

Means over seeds 8101-8103 on the same 1,067,211 rows. Fractional counts average integer seed outcomes. These are three fits of the same evaluation population.

Policy	Macro-F1	Exfil warnings	Missed warnings	Benign alerts	Acquire
Current	0.946050	99.698%	135.0	14448.7	0.000%
Always history	0.994053	99.772%	102.0	913.7	100.000%
Entropy	0.993970	99.772%	102.0	971.3	74.400%
Error-focused	0.946233	99.727%	122.0	14443.3	2.450%

Entropy selection has mean macro-F1 0.993970 versus 0.946233 for error-focused selection, exfiltration warning recall 99.772% versus 99.727%, and 971.3 versus 14,443.3 benign false alerts. History acquisition differs substantially: 74.400% versus 2.450%. These results compare the frozen policies with equal available budgets; they do not establish superiority at matched spending. The always-history reference attains macro-F1 0.994053 and 913.7 benign false alerts while retrieving history for every row.

Table 4-16. Separate AIT held-out executions.

Means across three fitting seeds, retaining distinct execution populations.

Execution	Policy	Macro-F1	Exfil warnings	Benign alerts
wilson	Current	0.993423	99.989%	55.7
wilson	Always history	0.994906	99.970%	52.7
wilson	Entropy	0.994800	99.970%	82.3
wilson	Error-focused	0.993678	99.989%	47.7
harrison	Current	0.909964	99.434%	14393.0
harrison	Always history	0.992345	99.592%	861.0
harrison	Entropy	0.992271	99.592%	889.0
harrison	Error-focused	0.910128	99.490%	14395.7

Wilson supplies a smaller score/warning tradeoff with the opposite policy ordering. Entropy has higher F1 than error-focused selection but misses one, three and eight additional exfiltration warnings for seeds 8101, 8102 and 8103. The mean difference is four of 21,219 exfiltration observations, or 0.018851 percentage points, with 34.7 additional benign false alerts. This is a secondary descriptive observation, not a replacement for the frozen primary direction. Its small absolute size limits practical significance. On Harrison, entropy improves both F1 and exfiltration warning recall; the two executions therefore do not support a uniform score/warning reversal.

Table 4-17. AIT paired sensitivity by execution and seed.

All deltas are error-focused minus entropy-focused selection. Positive missed-exfiltration counts mean more true exfiltration rows classified as benign.

Execution	Seed	Delta F1	Warning pp	Missed exfil	Benign alerts
wilson	8101	-0.001048	+0.0047	-1	-36
wilson	8102	-0.000982	+0.0141	-3	-36
wilson	8103	-0.001338	+0.0377	-8	-32
harrison	8101	-0.080391	-0.1577	+37	+13029
harrison	8102	-0.081340	-0.0512	+12	+13584
harrison	8103	-0.084697	-0.0981	+23	+13907

Giving each execution equal weight yields mean macro-F1 0.993536 and warning recall 99.781% for entropy, compared with 0.951903 and 99.739% for error-focused selection. Complete per-class precision, recall and F1, confusion matrices, unweighted and weighted errors, acquisition fractions and per-seed baselines are retained in the evidence and Appendix E. The 1,245-check audit passed with no failures.

In relation to RQ1, the separate source shows a small score/warning divergence in one held-out execution under the opposite policy ordering, while failing to reproduce the original primary direction. For RQ3, AIT qualifies for this declared three-class history task, not the unchanged four-class, two-group contract. RQ2 remains the original fixed-anchor temporal-composition experiment; the AIT extension does not re-estimate that effect.

Chapter 5: Discussion and Conclusions

5.1 Discussion of the Central Finding

A score is conditional on the records, target meanings, training composition, and decision rule used to calculate it. The controlled temporal experiment shows how large that dependence can be without changing the classifier family. The warning analysis shows that the aggregate score can move in a favorable direction while a consequential subset receives fewer attack labels. Together, the findings justify making those dependencies visible during model selection.

The practical contribution is a reporting and verification procedure. It can be used to review an internal model update, compare historical evidence features, or determine whether a proposed benchmark actually tests the intended deployment question. A research team gains an auditable reason to accept a limited claim, revise a protocol, or decline an unsupported comparison before purchasing more compute.

The AIT validation provides an empirical boundary on the policy result. Error-focused selection does not reproduce the original higher-F1/lower-warning direction on either held-out execution. Wilson nevertheless shows a small divergence in the opposite policy order, while Harrison favors entropy on both outcomes. This supports reading score, warning counts and false alerts jointly, without promoting a particular acquisition objective as universally superior. The large difference in realized acquisition rates also prevents a matched-spending superiority interpretation.

5.2 Contributions to the Body of Knowledge

The primary contribution is a controlled, reproducible measurement of how historical-evidence decisions affect stage recognition, attack-warning retention and

benign workload on matched evaluation records. The study makes the destination of errors explicit and retains comparisons that are favorable, unfavorable or unchanged. It also quantifies how much the headline warning-loss magnitude depends on a fitting seed.

The supporting temporal contribution fixes the anchor population and fitting class budgets while measuring a change in temporal access and training composition. The source-qualification contribution makes the eligibility of named releases executable: native-class support, timing and source dependence are checked before another model comparison is claimed.

These contributions add specific empirical evidence and reusable audit infrastructure to a literature that already recognizes temporal evaluation and attack-to-normal errors. They do not require a new detector or new mathematical metric. The research value is the evidence, controls and practical interpretability of the resulting evaluation procedure.

The new contribution additionally includes a frozen, external-source execution test on the complete AIT release. The retained negative primary result and small secondary tradeoff show where the earlier observation fails to generalize and where a narrower descriptive relationship remains visible. Shared laboratory construction, native label rules and the adapted evidence contract remain part of that contribution.

5.3 Applied Evaluation Procedure

For every consequential stage, a model evaluation should report exact-stage recall, warning recall, and their underlying counts on the same rows. It should include benign false alerts and the evaluated population size. The report should identify the

decision time, feature availability, class support in each split, source groups, and whether the evaluated source has already informed development.

This is a proposed reporting procedure, not a new industry standard or a mathematically new metric. The present evidence shows why the pair is useful in this setting. It does not prove that every omitted pair hides a loss or that every warning is actionable.

Table 5-1. Minimum audit record for a stage-classifier comparison.

Audit item	Required record	Decision it supports
Source and targets	Release hash, native labels, benign definition, label semantics	Whether the claim matches what was annotated
Timing	Training/evaluation boundaries, event completion, history availability	Whether the intended earlier-to-later comparison is possible
Dependence	Capture/run/campaign definitions, duplicate overlap, derivative sources	What constitutes independent evidence
Paired outcomes	Exact-stage and warning recall, wrong-stage and benign destinations	Whether stage improvement retains warnings
Workload	Benign false-alert count/rate, population, operating rule	Whether warning retention adds investigation burden
Uncertainty	Per-seed values, actual resampling units, unsupported draws	How much variation the available evidence describes
Reproducibility	Protocol/source commits, input/output hashes, all declared results	Whether another analyst can trace and recompute the claim

1. Define the operational decision and when its features become available. A full-flow detector should be evaluated as a completed-flow decision.
2. Inventory the actual release, target semantics, timestamps, and source relationships. Preserve uncertain fields explicitly.

3. Test native-class support and chronology before fitting. If the intended split is impossible, record that finding and register a different question separately.
4. Freeze the compared models, feature sets, operating rules, populations, and resource limits. Include simple controls.
5. Preserve row-linked predictions privately and calculate paired metrics from those same rows. Report all declared conditions.
6. Interpret stage accuracy, warning loss, and benign workload jointly. Use independent executions for a generalization claim and an appropriate calibration design for matched-workload claims.

The delivered planner, metric library, support diagnostic, and evidence receipts implement the corresponding parts of this workflow. They are a portable audit core, not a completed four-source fitting executor. The AIT adapter now implements a separate qualified three-class history contract; the original four-source contracts remain limited as reported. A generic tool cannot infer the meaning of a timestamp or turn a dataset name into an independent campaign.

5.4 Construct, Internal and External Validity

5.4.1 Construct validity

Author-stage labels are the targets. They are not independent attestations of successful host compromise, attacker intent, or stolen-file receipt. The movement target in the inspected UNRAVELED sensor concerns discovery. A non-benign prediction is called a warning for clarity, but no analyst workflow or downstream response was

measured. The paper consistently separates these measured proxies from stronger operational outcomes.

5.4.2 Internal validity

The temporal contrast fixes architecture, anchor, and class budgets but changes training composition and access to later observations. It does not isolate every distributional or dependence mechanism. Removing identical current-feature fingerprints addresses a specific overlap channel and leaves possible near-duplicate and workflow effects. The four-class source grouping and prior artifact preparation are part of the design.

History and acquisition interventions preserve targets, but wrong-host and missing-evidence conditions are synthetic perturbations. Their severity and frequency do not estimate real sensor failures. Prior acquisition costs, stage weights, and deadlines are illustrative choices. The reanalysis uses existing predictions and fixed decisions, so it cannot attribute a warning tradeoff to one mechanism by itself.

5.4.3 Statistical and external validity

The main campaign supplies only 35 movement evaluation rows and 18 in the temporal anchor. Five later captures contain correlated events from one workflow. Conditional capture-bootstrap intervals and fitting-seed variation cannot support population claims over organizations or APT campaigns. Missing-stage resamples require support-conditional interpretation, especially when a stage occurs in few captures.

The main data were already exposed during development. The warning-loss analysis was retrospective, and the complete paired reanalysis remains exploratory. Multiple stages and conditions are reported without a selective significance claim. The

extension audit does not produce independent replications by counting related or unavailable datasets.

5.4.4 Reproducibility and access limits

Public code, aggregate tables, support bounds, protocols, and hashes allow independent inspection of the analysis. Full recomputation requires the same qualified source artifacts and private row-linked prediction files, whose locations and hashes are recorded. The AIT reproduction archive additionally includes all eight unchanged publisher archives under CC BY 4.0 with attribution, prepared arrays, 42 model objects and every saved prediction. Source access, licensing, and authentication conditions can limit external reproduction. A hash establishes byte identity, not label correctness or authorization to redistribute data.

Independent computational checks establish consistency of specified calculations and source bindings. They are not a substitute for an external ground-truth review. No human annotation audit, deployment evaluation, ethics approval, or committee acceptance is represented as completed when it was not performed. No human subjects were recruited by these computational experiments.

The full-release extension removes the original file-coverage and class-cap restrictions for its own role-feature comparison. It retains one exposed campaign, sensor dependence and author-stage semantics. Its fixed four-class macro-F1 penalizes unsupported classes; comparisons across sensors with different class support are descriptive. Calibration-only review tolerances are illustrative, and a missing stage cannot be protected by an unobserved calibration measurement.

5.4.5 Scope of the finished manuscript

The completed manuscript makes a measurement claim on the available evidence and provides the full supporting artifact record. It does not wait for every desirable future study to report the results already obtained. The AIT extension adds two held-out laboratory executions for an adapted configuration. Their mixed findings broaden the empirical record while leaving exact four-class replication, unrelated-campaign generalization and operational benefit unresolved.

5.4.6 Validity of the external-execution extension

Whole-execution separation, final calendar separation, all-row fitting and strict prior-history construction strengthen the AIT test. They do not make two generated scenarios representative of independent real APT families. Native exfiltration labels are verified against the publisher's port/topology rule, not payload contents. The release aggregates probes without retaining sensor identity, so repeated observations may persist. Source taxonomies and evidence groups differ from PX081. The original-direction outcome was frozen before fitting, whereas the opposite-order Wilson interpretation is explicitly secondary and descriptive. Three seeds assess fitting sensitivity; no flow-level significance test or universal effect claim is made.

5.5 Recommendations for Future Research

The next empirical extension is an exact four-class, two-evidence-group replication on another qualified execution, followed by unrelated campaign families and measured collection constraints. The completed AIT test addresses disjoint laboratory executions for the adapted three-class history task. Its unreplicated primary direction and small secondary tradeoff make recurrence, effect size and source-label verification

explicit research questions. Additional architectures are useful only if they answer a defined comparison question; they are not a prerequisite for stating the completed measurement contribution.

An operational study could subsequently test whether the proposed joint report changes an analyst's or model-review team's decisions. Such a study would need actual users or deployment outcomes, a declared decision task and an appropriate comparison design. The present computational results do not stand in for that evidence.

The source intake checked Sandworm and CAM-LDS as possible extensions. Sandworm was previously evaluated in this project and has no native exfiltration flow class; CAM-LDS lacks simulated normal-user activity. A log-first annotation-linkage intake may support a different technique task, but those facts do not supply the missing full warning-and-benign-workload replication.

The September 26 follow-up acquired eight CRC-verified CAM-LDS archive members from two previously inspected executions and checked 3,332 flow records against four T1041 actions. Tool output reported successful downloads, but each action had multiple candidate C2 flows and the streams lacked verified legitimate-user labels. The complete publisher archive checksum was not verified. These additional bytes improve source inspection, yet do not create untouched executions or defensible benign/exfiltration flow labels. No unsupported replication model was fitted. The acquisition and qualification receipts accompany the final evidence.

5.6 Conclusions

This praxis provides an evidence-based evaluation method for exposing missed attack warnings that an improved APT classification score can conceal. In the observed

acquisition comparison, mean macro-F1 increased from 0.7148 to 0.7379 while exfiltration warning recall fell from 85.18% to 76.25%. The direction of the specified mean tradeoff survived every single-seed and single-capture omission, while its size remained strongly seed-sensitive.

Supporting experiments show that training composition changes the reported score on a fixed anchor, and that chronological history can improve both F1 and benign workload while still losing some stage-specific warnings. The dataset audit establishes which proposed extensions support the intended comparison and why the others do not.

The full-release extension accounts for all 6,877,157 released rows and evaluates the declared role-feature comparison with uncapped training. It strengthens coverage while exposing sensor-specific support limits; it supplies no independent campaign or operational validation.

The completed contribution is the controlled evidence and reproducible procedure: qualify the source, justify the temporal design, compare the same population when estimating a paired effect, separate wrong-stage warnings from attacks called benign, and report false alerts alongside both. The resulting claims are specific to the inspected evidence and ready for substantive scholarly review.

The AIT extension adds 3,465,342 inspected source rows, two later held-out executions and 42 fits under a separate frozen contract. The original named-policy direction does not replicate. A small opposite-order warning loss occurs on Wilson, and Harrison improves both outcomes. The defensible synthesis is therefore bounded: aggregate stage scores and warning retention can diverge on inspected records, but

neither the sign nor the size of a particular policy contrast is universal. A joint report makes these outcomes visible and preserves the adverse evidence.

References

- Bilot, T., Jiang, B., Li, Z., El Madhoun, N., Al Agha, K., Zouaoui, A., & Pasquier, T. (2025). Sometimes simpler is better: A comprehensive analysis of state-of-the-art provenance-based intrusion detection systems. In *34th USENIX Security Symposium* (pp. 7193-7212). USENIX Association.
<https://www.usenix.org/conference/usenixsecurity25/presentation/bilot>
- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1-26. <https://doi.org/10.1214/aos/1176344552>
- Guerra, L., Chapuis, T., Duc, G., Mozharovskyi, P., & Nguyen, V.-T. (2026). *How benchmarks and evaluation protocols shape conclusions in provenance-based intrusion detection* (Version 3) [Preprint]. arXiv.
<https://doi.org/10.48550/arXiv.2608.01454>
- Ha Thanh, D. (2026). A transfer-aware, deployment-oriented evaluation framework for NetFlow-based intrusion detection systems (TAN-IDS). *PLOS One*, 21(4), e0346801. <https://doi.org/10.1371/journal.pone.0346801>
- Ibrahim, N., Rajalakshmi, N. R., Sivakumar, V., & Sharmila, L. (2025). An optimized hybrid ensemble machine learning model combining multiple classifiers for detecting advanced persistent threats in networks. *Journal of Big Data*, 12, Article 212. <https://doi.org/10.1186/s40537-025-01272-w>
- Iturbe, E., Dalamagkas, C., Radoglou-Grammatikis, P., Rios, E., & Toledo, N. (2026). A pattern-aware LSTM-based approach for APT detection leveraging a realistic dataset for critical infrastructure security. *Future Generation Computer Systems*, 178, Article 108308. <https://doi.org/10.1016/j.future.2025.108308>
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems* (Vol. 30). Curran Associates.
<https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>
- Kilian, S., Viet Triem Tong, V., Lalande, J.-F., Majorczyk, F., Sanchez, A., Talon, N., Besson, P.-V., Orsini, H., Lledo, P., & Gimenez, P.-F. (2025). CasinoLimit: An offensive dataset labeled with MITRE ATT&CK techniques. In *2025 28th International Symposium on Research in Attacks, Intrusions and Defenses (RAID)*. IEEE. <https://doi.org/10.1109/RAID67961.2025.00039>
- Landauer, M., Hotwagner, W., Boenke, T., Skopik, F., & Wurzenberger, M. (2026). CAM-LDS: Cyber attack manifestations for automatic interpretation of system logs and security alerts. *International Journal of Information Security*, 25(5), Article 148. <https://doi.org/10.1007/s10207-026-01318-x>

- Landauer, M., Skopik, F., Frank, M., Hotwagner, W., Wurzenberger, M., & Rauber, A. (2023). Maintainable log datasets for evaluation of intrusion detection systems. *IEEE Transactions on Dependable and Secure Computing*, 20(4), 3466-3482. <https://arxiv.org/abs/2203.08580>
- Landauer, M., Skopik, F., Wurzenberger, M., Hotwagner, W., & Rauber, A. (2021). Have it your way: Generating customized log datasets with a model-driven simulation testbed. *IEEE Transactions on Reliability*, 70(1), 402-415. <https://doi.org/10.1109/TR.2020.3031317>
- LightGBM developers. (n.d.-a). *Features* (Version 4.6.0 documentation). Retrieved September 24, 2026, from <https://lightgbm.readthedocs.io/en/v4.6.0/Features.html>
- LightGBM developers. (n.d.-b). *lightgbm.LGBMClassifier* (Version 4.6.0 documentation). Retrieved September 24, 2026, from <https://lightgbm.readthedocs.io/en/v4.6.0/pythonapi/lightgbm.LGBMClassifier.html>
- LightGBM developers. (n.d.-c). *lightgbm.LGBMRegressor* (Version 4.6.0 documentation). Retrieved September 24, 2026, from <https://lightgbm.readthedocs.io/en/v4.6.0/pythonapi/lightgbm.LGBMRegressor.html>
- LightGBM developers. (n.d.-d). *Parameters* (Version 4.6.0 documentation). Retrieved September 24, 2026, from <https://lightgbm.readthedocs.io/en/v4.6.0/Parameters.html>
- Liu, J., Shen, Y., Simsek, M., Kantarci, B., Mouftah, H. T., Bagheri, M., & Djukic, P. (2022a). A new realistic benchmark for advanced persistent threats in network traffic. *IEEE Networking Letters*, 4(3), 162-166. <https://doi.org/10.1109/LNET.2022.3185553>
- Liu, J., Shen, Y., Simsek, M., Kantarci, B., Mouftah, H. T., Bagheri, M., & Djukic, P. (2022b). *SCVIC-APT-2021* [Dataset]. IEEE DataPort. <https://doi.org/10.21227/g2z5-ep97>
- Luengo Viñuela, M., Román-Gallego, J.-Á., Pérez-Delgado, M.-L., Conde, M. A., Vega-Hernández, M.-C., & Silva Varela, H. (2026). Detection of APTs by machine learning: A performance comparison. *Expert Systems*, 43(1), e70181. <https://doi.org/10.1111/exsy.70181>
- M K, V. P., S, S., H, A. G., Aggarwal, M., & V S, S. S. (2026). Advanced persistent threat detection through sequential analysis of network patterns with graph based learning approach. *Scientific Reports*, 16, Article 19998. <https://doi.org/10.1038/s41598-026-42756-w>

- Myneni, S., Chowdhary, A., Sabur, A., Sengupta, S., Agrawal, G., Huang, D., & Kang, M. (2020). DAPT 2020 - Constructing a benchmark dataset for advanced persistent threats. In G. Wang, A. Ciptadi, & A. Ahmadzadeh (Eds.), *Deployable machine learning for security defense (Communications in Computer and Information Science, Vol. 1271, pp. 138-163)*. Springer.
https://doi.org/10.1007/978-3-030-59621-7_8
- Myneni, S., Jha, K., Sabur, A., Agrawal, G., Deng, Y., Chowdhary, A., & Huang, D. (2023). Unraveled-A semi-synthetic dataset for advanced persistent threats. *Computer Networks*, 227, Article 109688.
<https://doi.org/10.1016/j.comnet.2023.109688>
- Othman, F. M., Mejri, M., & Alabdulatif, A. (2026). Temporal MITRE ATT&CK modelling for residual time-to-compromise estimation in multi-stage attacks. *Symmetry*, 18(9), Article 1439. <https://doi.org/10.3390/sym18091439>
- Pagan, G. (2026). *Advanced persistent threat event detection using graph machine learning* [Unpublished praxis manuscript]. The George Washington University.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85), 2825-2830. <https://jmlr.org/papers/v12/pedregosa11a.html>
- Pendlebury, F., Pierazzi, F., Jordaney, R., Kinder, J., & Cavallaro, L. (2019). TESSERACT: Eliminating experimental bias in malware classification across space and time. In *28th USENIX Security Symposium* (pp. 729-746). USENIX Association.
<https://www.usenix.org/conference/usenixsecurity19/presentation/pendlebury>
- Phan, T. V., & Bauschert, T. (2026). *Learning the APT kill chain: Temporal reasoning over provenance data for attack stage estimation* (Version 2) [Preprint]. arXiv.
<https://doi.org/10.48550/arXiv.2603.07560>
- Scikit-learn developers. (n.d.-a). *LogisticRegression* (Version 1.9 documentation, displayed as 1.9.1 at retrieval). Retrieved September 24, 2026, from https://scikit-learn.org/1.9/modules/generated/sklearn.linear_model.LogisticRegression.html
- Scikit-learn developers. (n.d.-b). *Ridge* (Version 1.7.2 documentation). Retrieved September 24, 2026, from https://scikit-learn.org/1.7/modules/generated/sklearn.linear_model.Ridge.html
- Shadabfar, H., Dehghan, M., & Sadeghian, B. (2025). DSRL-APT-2023: A new synthetic dataset for advanced persistent threats. *ISecure*, 17(2), 107-116.
<https://doi.org/10.22042/isecure.2025.214212>

- Soro, F., Landauer, M., Skopik, F., Hotwagner, W., & Wurzenberger, M. (2024). *AIT Netflow Data Set (Version 2)* [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.13168643>
- Tijjani, S. I., Ghita, B., Clarke, N., & Craven, M. (2026). *S-DAPT-2026: A stage-aware synthetic dataset for advanced persistent threat detection (Version 2)* [Withdrawn preprint]. arXiv. <https://doi.org/10.48550/arXiv.2601.06690>
- Uddin, M. A., Aryal, S., Bouadjenek, M. R., Al-Hawawreh, M., & Talukder, M. A. (2024). *Hierarchical classification for intrusion detection system: Effective design and empirical analysis* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2403.13013>
- Uddin, M. A., Aryal, S., Bouadjenek, M. R., Al-Hawawreh, M., & Talukder, M. A. (2025). Hierarchical classification for intrusion detection system: Effective design and empirical analysis. *Ad Hoc Networks*, 178, Article 103982. <https://doi.org/10.1016/j.adhoc.2025.103982>

Appendix A: Complete Results and Dataset Reporting

A.1 Complete Evaluation Inventory

The completed experiments support an applied measurement contribution: a higher aggregate APT-stage score does not necessarily preserve warnings for a particular attack stage. This result is reported alongside improvements, unsuccessful method comparisons, false-alert costs, and the limits of the source labels. This chapter adds a complete publication inventory to the main paired-results narrative; it introduces no new model fit, inference run, threshold, or favorable-condition selection.

Table A-1. Complete evaluation inventory

Completed study	Saved evaluations	Comparison coverage	Evaluation population
Context selection (PX080)	105	Seven arms $\times$ five conditions $\times$ three fits	208,094 UNRAVELED flows
Evidence acquisition (PX081)	171	Six policies $\times$ three budgets $\times$ three conditions $\times$ three fits, plus nine unrestricted references	Same 208,094 flows
Temporal composition (PX082)	18	Two feature views $\times$ three fitting protocols $\times$ three fits	Fixed anchor 104,051 flows; random comparison 210,226 flows
Secondary technique-policy transfer (PX083)	294	Seven arms $\times$ 21 views $\times$ two datasets	CasinoLimit: 920 targets; CAM-LDS: 4,209 targets

An evaluation record is one saved arm/condition/seed result, not a new dataset or independent attack. Primary fitting seeds reuse the same events. In the supplement, the 21 views represent 15 condition names: each of three random-loss conditions has three perturbations, and the remaining twelve conditions have one deterministic view. The supplement's underlying classifiers were previously trained on their respective sources;

only the two score selectors were fitted on CasinoLimit calibration predictions and applied unchanged to CAM-LDS.

The complete printable mean tables retain all **308 group means**. Machine-readable supplements contain all **588 evaluation records**, **1,764 class-metric records**, and **5,880 confusion cells**, plus **6,732 subgroup records** and **16,344 subgroup-class records** where the original experiments supplied them. Full source JSON preserves fields beyond the normalized tables. The acquisition reference-only arms have no matched acquisition budget and cannot support same-budget superiority claims.

A.2 All Reported Group Means

All 308 arm/condition/budget group means are retained. PX080-082 means use three fits on the same records. PX083 uses three perturbations for random-loss conditions and one deterministic view otherwise. Counts may therefore be fractional. No displayed group is an independent campaign.

Full per-fit precision, recall, F1, average precision, ROC-AUC, class supports, warning destinations and confusion counts are in the CSV tables. Blank ranking metrics mean unavailable; they are not replaced with invented estimates.

A.2.1 PX080

Table A-2. History-selection group means

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
clean	current_roles	0.7545	78.10% / 67.50%	67.60%	139.33
clean	context	0.7658	52.38% / 67.21%	67.64%	48.33

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
clean	fixed_fusion	0.7655	60.00% / 67.42%	67.54%	76.00
clean	confidence_gate	0.7650	60.00% / 67.42%	67.54%	77.00
clean	ordinary_gate	0.7659	64.76% / 67.36%	67.68%	85.67
clean	stage_harm_gate	0.7560	69.52% / 67.28%	67.81%	109.00
clean	context_dropout	0.7589	51.43% / 67.41%	67.83%	68.67
missing_half	current_roles	0.7545	78.10% / 67.50%	67.60%	139.33
missing_half	context	0.6626	29.52% / 66.99%	67.27%	28.67
missing_half	fixed_fusion	0.7602	44.76% / 67.33%	67.41%	54.33
missing_half	confidence_gate	0.7592	44.76% / 67.33%	67.40%	54.33
missing_half	ordinary_gate	0.7598	52.38% / 67.30%	67.52%	74.33
missing_half	stage_harm_gate	0.7561	60.95% / 67.17%	67.55%	93.67
missing_half	context_dropout	0.7595	62.86% / 67.43%	67.77%	95.00
missing_all	current_roles	0.7545	78.10% / 67.50%	67.60%	139.33
missing_all	context	0.4339	0.00% / 66.75%	66.75%	0.00
missing_all	fixed_fusion	0.7285	18.10% / 67.30%	67.30%	29.67
missing_all	confidence_gate	0.7265	18.10% / 67.29%	67.29%	29.67

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
missing_all	ordinary_gate	0.7419	36.19% / 67.23%	67.23%	62.33
missing_all	stage_harm_gate	0.7422	42.86% / 67.02%	67.11%	77.00
missing_all	context_dropout	0.7587	68.57% / 67.50%	67.64%	116.00
stale_5min	current_roles	0.7545	78.10% / 67.50%	67.60%	139.33
stale_5min	context	0.7261	21.90% / 67.22%	67.83%	37.67
stale_5min	fixed_fusion	0.7491	41.90% / 67.41%	67.56%	67.67
stale_5min	confidence_gate	0.7485	41.90% / 67.42%	67.57%	69.00
stale_5min	ordinary_gate	0.7456	49.52% / 67.30%	67.68%	85.00
stale_5min	stage_harm_gate	0.7472	63.81% / 67.33%	67.88%	113.33
stale_5min	context_dropout	0.7514	48.57% / 67.39%	67.88%	77.67
wrong_host	current_roles	0.7545	78.10% / 67.50%	67.60%	139.33
wrong_host	context	0.4559	0.00% / 66.80%	66.80%	0.00
wrong_host	fixed_fusion	0.7286	19.05% / 67.31%	67.31%	32.33
wrong_host	confidence_gate	0.7267	19.05% / 67.30%	67.30%	32.67
wrong_host	ordinary_gate	0.7406	35.24% / 67.24%	67.24%	61.67
wrong_host	stage_harm_gate	0.7430	44.76% / 67.03%	67.13%	81.33

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
wrong_host	context_dropout	0.7589	68.57% / 67.50%	67.64%	114.67

A.2.2 PX081

Table A-3. Evidence-acquisition group means

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
clean / B1	none	0.7523	78.10% / 67.52%	67.52%	164.00
clean / B1	roles_first	0.7136	85.71% / 67.52%	88.34%	179.67
clean / B1	history_first	0.7136	85.71% / 67.52%	88.34%	179.67
clean / B1	random	0.7136	85.71% / 67.52%	88.34%	179.67
clean / B1	entropy	0.7139	81.90% / 67.52%	87.95%	174.00
clean / B1	harm	0.7500	81.90% / 67.52%	67.83%	169.67
clean / B2	none	0.7523	78.10% / 67.52%	67.52%	164.00
clean / B2	roles_first	0.7136	85.71% / 67.52%	88.34%	179.67
clean / B2	history_first	0.7594	58.10% / 67.13%	67.17%	86.67
clean / B2	random	0.7213	76.19% / 67.34%	77.75%	133.67
clean / B2	entropy	0.7181	63.81% / 67.29%	87.58%	101.33
clean / B2	harm	0.7551	61.90% / 67.25%	67.33%	107.33

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
clean / B3	none	0.7523	78.10% / 67.52%	67.52%	164.00
clean / B3	roles_first	0.7153	68.57% / 67.15%	85.20%	115.33
clean / B3	history_first	0.7153	68.57% / 67.15%	85.20%	115.33
clean / B3	random	0.7153	68.57% / 67.15%	85.20%	115.33
clean / B3	entropy	0.7148	65.71% / 67.29%	85.18%	111.33
clean / B3	harm	0.7379	69.52% / 67.18%	76.25%	122.33
clean	full_context_reference (unrestricted reference)	0.7153	68.57% / 67.15%	85.20%	115.33
delayed_unavailable / B1	none	0.7523	78.10% / 67.52%	67.52%	164.00
delayed_unavailable / B1	roles_first	0.7167	83.81% / 67.52%	81.81%	173.33
delayed_unavailable / B1	history_first	0.7167	83.81% / 67.52%	81.81%	173.33
delayed_unavailable / B1	random	0.7167	83.81% / 67.52%	81.81%	173.33
delayed_unavailable / B1	entropy	0.7168	80.95% / 67.52%	81.54%	169.67
delayed_unavailable / B1	harm	0.7509	80.95% / 67.52%	67.73%	167.67
delayed_unavailable / B2	none	0.7523	78.10% / 67.52%	67.52%	164.00
delayed_unavailable / B2	roles_first	0.7167	83.81% / 67.52%	81.81%	173.33

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
delayed_unavailable / B2	history_first	0.7547	71.43% / 67.39%	67.41%	135.00
delayed_unavailable / B2	random	0.7223	77.14% / 67.46%	74.73%	156.67
delayed_unavailable / B2	entropy	0.7187	76.19% / 67.44%	81.41%	143.33
delayed_unavailable / B2	harm	0.7544	74.29% / 67.42%	67.46%	144.33
delayed_unavailable / B3	none	0.7523	78.10% / 67.52%	67.52%	164.00
delayed_unavailable / B3	roles_first	0.7172	80.00% / 67.48%	81.28%	160.00
delayed_unavailable / B3	history_first	0.7273	75.24% / 67.41%	72.90%	141.67
delayed_unavailable / B3	random	0.7202	78.10% / 67.44%	77.20%	153.33
delayed_unavailable / B3	entropy	0.7180	75.24% / 67.43%	81.00%	144.00
delayed_unavailable / B3	harm	0.7427	75.24% / 67.43%	69.20%	146.33
delayed_unavailable	full_context_reference (unrestricted reference)	0.7153	68.57% / 67.15%	85.20%	115.33
wrong_host_history / B1	none	0.7523	78.10% / 67.52%	67.52%	164.00
wrong_host_history / B1	roles_first	0.7136	85.71% / 67.52%	88.34%	179.67
wrong_host_history / B1	history_first	0.7136	85.71% / 67.52%	88.34%	179.67
wrong_host_history / B1	random	0.7136	85.71% / 67.52%	88.34%	179.67

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
wrong_host_history / B1	entropy	0.7139	81.90% / 67.52%	87.95%	174.00
wrong_host_history / B1	harm	0.7500	81.90% / 67.52%	67.83%	169.67
wrong_host_history / B2	none	0.7523	78.10% / 67.52%	67.52%	164.00
wrong_host_history / B2	roles_first	0.7136	85.71% / 67.52%	88.34%	179.67
wrong_host_history / B2	history_first	0.5449	23.81% / 66.89%	66.89%	39.67
wrong_host_history / B2	random	0.6577	58.10% / 67.22%	77.61%	110.33
wrong_host_history / B2	entropy	0.5687	40.00% / 67.20%	87.46%	66.00
wrong_host_history / B2	harm	0.6921	40.95% / 67.08%	67.13%	81.00
wrong_host_history / B3	none	0.7523	78.10% / 67.52%	67.52%	164.00
wrong_host_history / B3	roles_first	0.6793	43.81% / 66.96%	77.38%	74.00
wrong_host_history / B3	history_first	0.6793	43.81% / 66.96%	77.38%	74.00
wrong_host_history / B3	random	0.6793	43.81% / 66.96%	77.38%	74.00
wrong_host_history / B3	entropy	0.6873	40.00% / 67.22%	77.52%	73.00
wrong_host_history / B3	harm	0.6763	43.81% / 67.01%	77.37%	87.00
wrong_host_history	full_context_reference (unrestricted reference)	0.6793	43.81% / 66.96%	77.38%	74.00

A.2.3 PX082

Table A-4. Temporal-comparison group means

Condition / view / budget	Arm	Macro-F1	Exact recall: movement / exfil.	Exfil. warning recall	Benign flags
current	past_only_anchor	0.7365	25.93% / 67.25%	67.27%	24.00
current	time_mixed_anchor	0.7997	61.11% / 98.72%	99.28%	88.67
current	conventional_random	0.8093	57.84% / 92.15%	99.74%	84.00
current_history	past_only_anchor	0.7582	29.63% / 66.40%	66.86%	14.33
current_history	time_mixed_anchor	0.7964	57.41% / 84.59%	84.71%	47.67
current_history	conventional_random	0.8507	64.71% / 97.48%	99.72%	49.33

A.2.4 PX083

A.2.5 casino

Table A-5. CasinoLimit policy group means

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
clean	current	0.7368	82.35%	0.8674 / 0.9607	7.00
clean	context	0.5957	82.35%	0.8522 / 0.9762	16.00
clean	fixed_fusion	0.7179	82.35%	0.8606 / 0.9711	8.00
clean	mixed_dropout	0.5833	82.35%	0.8438 / 0.9693	17.00
clean	confidence_gate	0.7179	82.35%	0.8717 / 0.9684	8.00
clean	ordinary_gate	0.7368	82.35%	0.8672 / 0.9682	7.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
clean	target_cost_gate	0.5957	82.35%	0.8580 / 0.9667	16.00
random_25	current	0.3996	86.27%	0.7923 / 0.9622	41.67
random_25	context	0.5271	82.35%	0.8028 / 0.9714	22.33
random_25	fixed_fusion	0.4665	84.31%	0.8057 / 0.9678	30.33
random_25	mixed_dropout	0.5468	82.35%	0.8109 / 0.9623	20.67
random_25	confidence_gate	0.4665	84.31%	0.7988 / 0.9652	30.33
random_25	ordinary_gate	0.3996	86.27%	0.8025 / 0.9651	41.67
random_25	target_cost_gate	0.5271	82.35%	0.8055 / 0.9647	22.33
random_50	current	0.2368	88.24%	0.6548 / 0.9468	94.67
random_50	context	0.3274	80.39%	0.7210 / 0.9545	53.00
random_50	fixed_fusion	0.2768	84.31%	0.6934 / 0.9511	72.33
random_50	mixed_dropout	0.3649	78.43%	0.7728 / 0.9556	43.00
random_50	confidence_gate	0.2768	84.31%	0.6700 / 0.9494	72.33
random_50	ordinary_gate	0.2368	88.24%	0.7181 / 0.9520	94.67
random_50	target_cost_gate	0.3274	80.39%	0.7211 / 0.9512	53.00
random_75	current	0.1361	72.55%	0.5075 / 0.8203	150.67
random_75	context	0.1946	68.63%	0.5781 / 0.8252	91.00
random_75	fixed_fusion	0.1678	70.59%	0.5482 / 0.8238	114.33
random_75	mixed_dropout	0.2084	62.75%	0.5962 / 0.8339	75.00
random_75	confidence_gate	0.1678	70.59%	0.5257 / 0.8217	114.33
random_75	ordinary_gate	0.1361	72.55%	0.5753 / 0.8255	150.67

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
random_75	target_cost_gate	0.1946	68.63%	0.5783 / 0.8256	91.00
support_burst_60	current	0.7368	82.35%	0.8674 / 0.9607	7.00
support_burst_60	context	0.5769	88.24%	0.8500 / 0.9647	20.00
support_burst_60	fixed_fusion	0.6818	88.24%	0.8644 / 0.9650	12.00
support_burst_60	mixed_dropout	0.4667	82.35%	0.8161 / 0.9565	29.00
support_burst_60	confidence_gate	0.6818	88.24%	0.8581 / 0.9616	12.00
support_burst_60	ordinary_gate	0.7368	82.35%	0.8630 / 0.9618	7.00
support_burst_60	target_cost_gate	0.5769	88.24%	0.8518 / 0.9591	20.00
command_records_absent	current	0.2258	82.35%	0.6714 / 0.9414	93.00
command_records_absent	context	0.3607	64.71%	0.6319 / 0.9479	33.00
command_records_absent	fixed_fusion	0.4138	70.59%	0.6424 / 0.9510	29.00
command_records_absent	mixed_dropout	0.2121	82.35%	0.6939 / 0.9415	101.00
command_records_absent	confidence_gate	0.4138	70.59%	0.6722 / 0.9496	29.00
command_records_absent	ordinary_gate	0.2258	82.35%	0.6368 / 0.9506	93.00
command_records_absent	target_cost_gate	0.3607	64.71%	0.6339 / 0.9480	33.00
delay_30_deadline_0	current	0.2258	82.35%	0.6714 / 0.9414	93.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
delay_30_deadline_0	context	0.3729	64.71%	0.6224 / 0.9505	31.00
delay_30_deadline_0	fixed_fusion	0.4364	70.59%	0.6427 / 0.9525	26.00
delay_30_deadline_0	mixed_dropout	0.2205	82.35%	0.6919 / 0.9426	96.00
delay_30_deadline_0	confidence_gate	0.4364	70.59%	0.6499 / 0.9506	26.00
delay_30_deadline_0	ordinary_gate	0.2258	82.35%	0.6369 / 0.9511	93.00
delay_30_deadline_0	target_cost_gate	0.3729	64.71%	0.6241 / 0.9492	31.00
delay_30_deadline_30	current	0.7368	82.35%	0.8674 / 0.9607	7.00
delay_30_deadline_30	context	0.5957	82.35%	0.8522 / 0.9762	16.00
delay_30_deadline_30	fixed_fusion	0.7179	82.35%	0.8606 / 0.9711	8.00
delay_30_deadline_30	mixed_dropout	0.5833	82.35%	0.8438 / 0.9693	17.00
delay_30_deadline_30	confidence_gate	0.7179	82.35%	0.8717 / 0.9684	8.00
delay_30_deadline_30	ordinary_gate	0.7368	82.35%	0.8672 / 0.9682	7.00
delay_30_deadline_30	target_cost_gate	0.5957	82.35%	0.8580 / 0.9667	16.00
delay_120_deadline_0	current	0.2258	82.35%	0.6714 / 0.9414	93.00
delay_120_deadline_0	context	0.3607	64.71%	0.6319 / 0.9479	33.00
delay_120_deadline_0	fixed_fusion	0.4138	70.59%	0.6424 / 0.9510	29.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
delay_120_deadline_0	mixed_dropout	0.2121	82.35%	0.6939 / 0.9415	101.00
delay_120_deadline_0	confidence_gate	0.4138	70.59%	0.6722 / 0.9496	29.00
delay_120_deadline_0	ordinary_gate	0.2258	82.35%	0.6368 / 0.9506	93.00
delay_120_deadline_0	target_cost_gate	0.3607	64.71%	0.6339 / 0.9480	33.00
delay_120_deadline_30	current	0.2258	82.35%	0.6714 / 0.9414	93.00
delay_120_deadline_30	context	0.3607	64.71%	0.6335 / 0.9495	33.00
delay_120_deadline_30	fixed_fusion	0.4138	70.59%	0.6428 / 0.9522	29.00
delay_120_deadline_30	mixed_dropout	0.2105	82.35%	0.6958 / 0.9423	102.00
delay_120_deadline_30	confidence_gate	0.4138	70.59%	0.6732 / 0.9500	29.00
delay_120_deadline_30	ordinary_gate	0.2258	82.35%	0.6368 / 0.9507	93.00
delay_120_deadline_30	target_cost_gate	0.3607	64.71%	0.6349 / 0.9484	33.00
delay_120_deadline_120	current	0.7368	82.35%	0.8674 / 0.9607	7.00
delay_120_deadline_120	context	0.5957	82.35%	0.8522 / 0.9762	16.00
delay_120_deadline_120	fixed_fusion	0.7179	82.35%	0.8606 / 0.9711	8.00
delay_120_deadline_120	mixed_dropout	0.5833	82.35%	0.8438 / 0.9693	17.00
delay_120_deadline_120	confidence_gate	0.7179	82.35%	0.8717 / 0.9684	8.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
delay_120_deadline_120	ordinary_gate	0.7368	82.35%	0.8672 / 0.9682	7.00
delay_120_deadline_120	target_cost_gate	0.5957	82.35%	0.8580 / 0.9667	16.00
execve_absent	current	0.3226	88.24%	0.8508 / 0.9724	61.00
execve_absent	context	0.3704	88.24%	0.8408 / 0.9836	49.00
execve_absent	fixed_fusion	0.4000	88.24%	0.8556 / 0.9785	43.00
execve_absent	mixed_dropout	0.5091	82.35%	0.8138 / 0.9653	24.00
execve_absent	confidence_gate	0.4000	88.24%	0.8526 / 0.9780	43.00
execve_absent	ordinary_gate	0.3226	88.24%	0.8530 / 0.9774	61.00
execve_absent	target_cost_gate	0.3704	88.24%	0.8388 / 0.9775	49.00
proctitle_absent	current	0.7778	82.35%	0.8081 / 0.9476	5.00
proctitle_absent	context	0.7179	82.35%	0.7788 / 0.9666	8.00
proctitle_absent	fixed_fusion	0.7778	82.35%	0.7816 / 0.9616	5.00
proctitle_absent	mixed_dropout	0.3784	82.35%	0.7978 / 0.9542	43.00
proctitle_absent	confidence_gate	0.7778	82.35%	0.8230 / 0.9597	5.00
proctitle_absent	ordinary_gate	0.7778	82.35%	0.7840 / 0.9592	5.00
proctitle_absent	target_cost_gate	0.7179	82.35%	0.7852 / 0.9575	8.00
syscall_absent	current	0.3636	94.12%	0.8738 / 0.9505	55.00
syscall_absent	context	0.6522	88.24%	0.8762 / 0.9488	14.00
syscall_absent	fixed_fusion	0.5263	88.24%	0.8760 / 0.9495	25.00
syscall_absent	mixed_dropout	0.8000	82.35%	0.8748 / 0.9543	4.00
syscall_absent	confidence_gate	0.5263	88.24%	0.8695 / 0.9487	25.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
syscall_absent	ordinary_gate	0.3636	94.12%	0.8701 / 0.9487	55.00
syscall_absent	target_cost_gate	0.6522	88.24%	0.8765 / 0.9486	14.00
path_absent	current	0.7692	88.24%	0.8712 / 0.9558	7.00
path_absent	context	0.5926	94.12%	0.8698 / 0.9761	21.00
path_absent	fixed_fusion	0.6522	88.24%	0.8665 / 0.9696	14.00
path_absent	mixed_dropout	0.4918	88.24%	0.8559 / 0.9764	29.00
path_absent	confidence_gate	0.6522	88.24%	0.8626 / 0.9566	14.00
path_absent	ordinary_gate	0.7692	88.24%	0.8548 / 0.9563	7.00
path_absent	target_cost_gate	0.5926	94.12%	0.8685 / 0.9570	21.00

A.2.6 camlds

Table A-6. CAM-LDS policy group means

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
clean	current	0.0495	87.00%	0.0238 / 0.5087	3329.00
clean	context	0.0534	88.00%	0.0277 / 0.5595	3108.00
clean	fixed_fusion	0.0524	88.00%	0.0249 / 0.5381	3171.00
clean	mixed_dropout	0.0527	85.00%	0.0276 / 0.5660	3038.00
clean	confidence_gate	0.0524	88.00%	0.0250 / 0.5348	3171.00
clean	ordinary_gate	0.0495	87.00%	0.0234 / 0.5070	3325.00
clean	target_cost_gate	0.0534	88.00%	0.0277 / 0.5593	3108.00
random_25	current	0.0454	49.33%	0.0241 / 0.5064	2023.33
random_25	context	0.0468	61.67%	0.0290 / 0.5239	2474.33

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
random_25	fixed_fusion	0.0475	59.00%	0.0259 / 0.5159	2323.00
random_25	mixed_dropout	0.0515	81.00%	0.0259 / 0.5317	2967.33
random_25	confidence_gate	0.0475	59.00%	0.0286 / 0.5209	2323.00
random_25	ordinary_gate	0.0455	49.33%	0.0248 / 0.5164	2020.00
random_25	target_cost_gate	0.0468	61.67%	0.0290 / 0.5242	2474.33
random_50	current	0.0452	28.00%	0.0248 / 0.5101	1112.33
random_50	context	0.0469	51.00%	0.0296 / 0.5277	2022.67
random_50	fixed_fusion	0.0472	45.00%	0.0300 / 0.5214	1760.67
random_50	mixed_dropout	0.0491	71.00%	0.0262 / 0.5279	2724.00
random_50	confidence_gate	0.0472	45.00%	0.0293 / 0.5246	1760.67
random_50	ordinary_gate	0.0452	28.00%	0.0256 / 0.5175	1111.00
random_50	target_cost_gate	0.0469	51.00%	0.0295 / 0.5257	2022.67
random_75	current	0.0493	15.00%	0.0250 / 0.5040	493.00
random_75	context	0.0446	32.33%	0.0254 / 0.4976	1318.33
random_75	fixed_fusion	0.0427	24.33%	0.0249 / 0.4999	1013.67
random_75	mixed_dropout	0.0467	48.33%	0.0242 / 0.4964	1920.33
random_75	confidence_gate	0.0427	24.33%	0.0253 / 0.4980	1013.67
random_75	ordinary_gate	0.0493	15.00%	0.0253 / 0.5053	493.00
random_75	target_cost_gate	0.0446	32.33%	0.0254 / 0.4987	1318.33
support_burst_60	current	0.0495	87.00%	0.0238 / 0.5087	3329.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
support_burst_60	context	0.0490	95.00%	0.0223 / 0.4862	3685.00
support_burst_60	fixed_fusion	0.0500	92.00%	0.0224 / 0.4956	3491.00
support_burst_60	mixed_dropout	0.0488	91.00%	0.0216 / 0.4811	3541.00
support_burst_60	confidence_gate	0.0500	92.00%	0.0224 / 0.4907	3491.00
support_burst_60	ordinary_gate	0.0495	87.00%	0.0238 / 0.5079	3326.00
support_burst_60	target_cost_gate	0.0490	95.00%	0.0223 / 0.4862	3685.00
command_records_absent	current	0.0000	0.00%	0.0241 / 0.5046	2.00
command_records_absent	context	0.0384	15.00%	0.0250 / 0.5338	666.00
command_records_absent	fixed_fusion	0.0000	0.00%	0.0245 / 0.5254	4.00
command_records_absent	mixed_dropout	0.0532	84.00%	0.0256 / 0.5444	2976.00
command_records_absent	confidence_gate	0.0000	0.00%	0.0242 / 0.5293	4.00
command_records_absent	ordinary_gate	0.0000	0.00%	0.0253 / 0.5378	2.00
command_records_absent	target_cost_gate	0.0384	15.00%	0.0247 / 0.5305	666.00
delay_30_deadline_0	current	0.0000	0.00%	0.0241 / 0.5046	2.00
delay_30_deadline_0	context	0.0319	4.00%	0.0249 / 0.5306	147.00
delay_30_deadline_0	fixed_fusion	0.0000	0.00%	0.0238 / 0.5191	13.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
delay_30_deadline_0	mixed_dropout	0.0518	79.00%	0.0289 / 0.5706	2870.00
delay_30_deadline_0	confidence_gate	0.0000	0.00%	0.0244 / 0.5241	13.00
delay_30_deadline_0	ordinary_gate	0.0000	0.00%	0.0247 / 0.5269	2.00
delay_30_deadline_0	target_cost_gate	0.0319	4.00%	0.0252 / 0.5280	147.00
delay_30_deadline_30	current	0.0495	87.00%	0.0238 / 0.5087	3329.00
delay_30_deadline_30	context	0.0534	88.00%	0.0277 / 0.5595	3108.00
delay_30_deadline_30	fixed_fusion	0.0524	88.00%	0.0249 / 0.5381	3171.00
delay_30_deadline_30	mixed_dropout	0.0527	85.00%	0.0276 / 0.5660	3038.00
delay_30_deadline_30	confidence_gate	0.0524	88.00%	0.0250 / 0.5348	3171.00
delay_30_deadline_30	ordinary_gate	0.0495	87.00%	0.0234 / 0.5070	3325.00
delay_30_deadline_30	target_cost_gate	0.0534	88.00%	0.0277 / 0.5593	3108.00
delay_120_deadline_0	current	0.0000	0.00%	0.0241 / 0.5046	2.00
delay_120_deadline_0	context	0.0384	15.00%	0.0250 / 0.5338	666.00
delay_120_deadline_0	fixed_fusion	0.0000	0.00%	0.0245 / 0.5254	4.00
delay_120_deadline_0	mixed_dropout	0.0532	84.00%	0.0256 / 0.5444	2976.00
delay_120_deadline_0	confidence_gate	0.0000	0.00%	0.0242 / 0.5293	4.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
delay_120_deadline_0	ordinary_gate	0.0000	0.00%	0.0253 / 0.5378	2.00
delay_120_deadline_0	target_cost_gate	0.0384	15.00%	0.0247 / 0.5305	666.00
delay_120_deadline_30	current	0.0000	0.00%	0.0241 / 0.5046	2.00
delay_120_deadline_30	context	0.0331	8.00%	0.0247 / 0.5294	376.00
delay_120_deadline_30	fixed_fusion	0.0000	0.00%	0.0242 / 0.5211	6.00
delay_120_deadline_30	mixed_dropout	0.0529	86.00%	0.0259 / 0.5443	3063.00
delay_120_deadline_30	confidence_gate	0.0000	0.00%	0.0244 / 0.5272	6.00
delay_120_deadline_30	ordinary_gate	0.0000	0.00%	0.0249 / 0.5315	2.00
delay_120_deadline_30	target_cost_gate	0.0331	8.00%	0.0242 / 0.5241	376.00
delay_120_deadline_120	current	0.0495	87.00%	0.0238 / 0.5087	3329.00
delay_120_deadline_120	context	0.0534	88.00%	0.0277 / 0.5595	3108.00
delay_120_deadline_120	fixed_fusion	0.0524	88.00%	0.0249 / 0.5381	3171.00
delay_120_deadline_120	mixed_dropout	0.0527	85.00%	0.0276 / 0.5660	3038.00
delay_120_deadline_120	confidence_gate	0.0524	88.00%	0.0250 / 0.5348	3171.00
delay_120_deadline_120	ordinary_gate	0.0495	87.00%	0.0234 / 0.5070	3325.00
delay_120_deadline_120	target_cost_gate	0.0534	88.00%	0.0277 / 0.5593	3108.00
execve_absent	current	0.0478	84.00%	0.0243 / 0.5096	3327.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
execve_absent	context	0.0514	74.00%	0.0257 / 0.5388	2707.00
execve_absent	fixed_fusion	0.0506	77.00%	0.0245 / 0.5227	2865.00
execve_absent	mixed_dropout	0.0525	85.00%	0.0294 / 0.5795	3053.00
execve_absent	confidence_gate	0.0506	77.00%	0.0248 / 0.5272	2865.00
execve_absent	ordinary_gate	0.0479	84.00%	0.0235 / 0.5023	3321.00
execve_absent	target_cost_gate	0.0514	74.00%	0.0256 / 0.5343	2707.00
proctitle_absent	current	0.0075	1.00%	0.0237 / 0.5038	165.00
proctitle_absent	context	0.0406	22.00%	0.0243 / 0.5299	962.00
proctitle_absent	fixed_fusion	0.0000	0.00%	0.0243 / 0.5294	27.00
proctitle_absent	mixed_dropout	0.0515	85.00%	0.0255 / 0.5381	3116.00
proctitle_absent	confidence_gate	0.0000	0.00%	0.0252 / 0.5366	27.00
proctitle_absent	ordinary_gate	0.0075	1.00%	0.0245 / 0.5282	165.00
proctitle_absent	target_cost_gate	0.0406	22.00%	0.0243 / 0.5308	962.00
syscall_absent	current	0.0404	9.00%	0.0251 / 0.5275	336.00
syscall_absent	context	0.0445	53.00%	0.0276 / 0.5144	2230.00
syscall_absent	fixed_fusion	0.0457	52.00%	0.0247 / 0.5172	2122.00
syscall_absent	mixed_dropout	0.0510	85.00%	0.0246 / 0.5189	3151.00
syscall_absent	confidence_gate	0.0457	52.00%	0.0279 / 0.5247	2122.00
syscall_absent	ordinary_gate	0.0404	9.00%	0.0249 / 0.5224	336.00
syscall_absent	target_cost_gate	0.0445	53.00%	0.0277 / 0.5179	2230.00
path_absent	current	0.0494	96.00%	0.0275 / 0.5628	3692.00

Condition	Arm	T1105 F1	Recall	AP / ROC-AUC	Other-label flags
path_absent	context	0.0506	94.00%	0.0293 / 0.5723	3525.00
path_absent	fixed_fusion	0.0494	94.00%	0.0306 / 0.5783	3610.00
path_absent	mixed_dropout	0.0499	76.00%	0.0276 / 0.5488	2868.00
path_absent	confidence_gate	0.0494	94.00%	0.0288 / 0.5666	3610.00
path_absent	ordinary_gate	0.0495	96.00%	0.0277 / 0.5661	3684.00
path_absent	target_cost_gate	0.0506	94.00%	0.0293 / 0.5724	3525.00

A.3 Stage Metrics and Source-Record Access

The complete supplement retains the per-class precision, recall, F1, ROC-AUC and average precision fields reported by each source, with confusion matrices and subgroup records. Blank normalized CSV fields represent metrics not supplied by the original experiment; they are not zeros or reconstructed ranking scores. Every group mean printed above is backed by its component evaluation records. The source JSON preserves additional nested fields and original null values.

Readers should compare only compatible populations, targets and operating decisions. Primary stage comparisons and secondary T1105 policy views have different units and denominators. The saved tables provide the full inventory without converting related views into independent empirical replications.

Appendix B: Reproducibility, Provenance and Evidence Access

B.1 Experimental Parameters and Source Provenance

Table B-1. Experimental settings and fitting seeds

Study	Boosting iterations / leaves	Fitting seeds	Key fixed choices
History selection	180 / 15	20260924-20260926	4 forward folds; target weights 1/1/4/4 for weighted selector
Evidence acquisition	150 / 15	8101-8103	2 optional groups; costs 1/2; budgets 1/2/3; deadline 1
Temporal comparison	200 / 15	20260923-20260925	Same anchor and class budgets; earlier/mixed/random arms
Paired reanalysis	No fits	Bootstrap 20260923	2,000 whole-capture draws shared across fitting seeds

All main fitted models use LightGBM with fixed settings; no model family is selected after seeing the evaluation scores. The reanalysis fits zero models. D1's prospective Random Forest control belongs to unexecuted qualified-source comparisons and is not listed as an observed result.

Table B-2. Source freezes and evidence bindings

Artifact	Before-execution commit	Evidence
History-selection protocol/source	06c5037	Original model audit
Acquisition and temporal protocol/source	2da1a1c	Original model audits
D1 support diagnostic	ea9956c	Native-class necessary interval
Independent source-count verifier	e6b5799	Direct timestamp-block sweep
Retrospective paired reanalysis	c0d884e	Saved-prediction comparison; exploratory

The prepared UNRAVELED SHA-256 is
b2a491474e722f4dabcd4c419c83a4a6b49f08dfc3bc059aa42ef2aaa4c3de14. Exact
prediction hashes, source versions, and code bindings are in the linked receipts. The main
recorded numerical environment is NumPy 2.2.6, LightGBM 4.6.0, and scikit-learn 1.7.2;
current reanalysis and document build versions are recorded separately.

B.2 Claim Traceability

Table B-3. Claim traceability

Claim	Measured support	Evidence location
Temporal score sensitivity	+0.0632 macro-F1; +35.19 pp movement recall	Temporal report
Metric/warning tradeoff	Clean budget 3: +0.0231 macro-F1; -8.93 pp warning recall	Paired report
Chronological history gains	Macro-F1 .7365 to .7582; alerts 24.0 to 14.3	Original means
Native-class cutoff limitation	DAPT lower bound exceeds upper bound	Independent verification
Source qualification	SCVIC uncertainty; DSRL dependency; S-DAPT unavailable	Qualification report
Literature differentiation	Existing metrics; controlled measurement contribution	Literature audit

The machine-readable result tables preserve all reported fitting seeds, budgets, conditions, stages, and bootstrap support counts. The paper's compact tables summarize these records. Zero-change and unfavorable results remain available. The evidence index maps each artifact to the calculation it supports and distinguishes public aggregates from private row-level inputs.

B.3 Public Reproduction and Access Limits

The original evidence archive was extracted into a new directory and its public arithmetic verifier was run with Python 3.11.9 and NumPy 2.2.6 in a fresh virtual environment. All 36 paired comparisons and 720 reported metric intervals were reproduced from 66 public aggregate tables and the 2,000-draw resampling plan. All 206 original manifest hashes matched. Python file-open auditing recorded no access outside the allowed extracted files, new output/runtime directories and base Python runtime. This is a fresh local-runtime check, not an operating-system sandbox, another laboratory's replication or a refit from raw traces.

The initial package check exposed one missing relative link: the original README linked to the archive itself, which was not inside that archive. Adding the byte-identical downloaded archive at the linked location allowed the unchanged package verifier to pass all 74 local links. No original extracted member changed. Both the initial failure and repaired pass are retained in the clean-extraction report. The complete review bundle includes the original archive alongside its extracted contents, so the original verification commands work after extraction.

These checks verify aggregate arithmetic and artifact bindings. They cannot independently reconstruct row identities, validate attack labels, establish successful exfiltration or reproduce private-source model fitting. The original access and inference limits remain applicable.

B.4 Document and Reference Provenance

The five-chapter organization, title-page author information and What/Why/How GMR convention follow the author's original GWU GML praxis. The current GWU

online-program 2026 template and official formatting requirements were also inspected. The reference record identifies the source files and hashes. The original document's certification and dates are historical text; they are not carried forward as approval of this study.

The reference review is targeted and uses primary sources, author manuscripts and publisher metadata where available. Some final publisher full texts were inaccessible; the cited version and access limits are recorded in the literature audit. Bibliographic entries identify inspected preprints separately from final publications. The dataset papers establish provenance, while the local source inspections establish what the acquired artifacts support.

AI-assisted agents were used for code development, computational checks, literature triage, document assembly and layout review. Computational agreement and same-team review do not substitute for the author's responsibility to inspect the research, for independent label validation or for adviser and institutional review. No external peer review or institutional approval is represented as completed.

Appendix C: Original GWU GML GMR Reference

The following image is the exact Graphical Model of Research extracted from the author's earlier manuscript, *Advanced Persistent Threat Event Detection Using Graph Machine Learning* (Pagan, 2026). It is reproduced as a historical formatting and research-structure reference. Its graph models, Optuna tuning, AWS acquisition and evaluation statements describe that earlier document and are not the methods or results of this praxis. The adapted current-study GMR appears in Chapter 3.

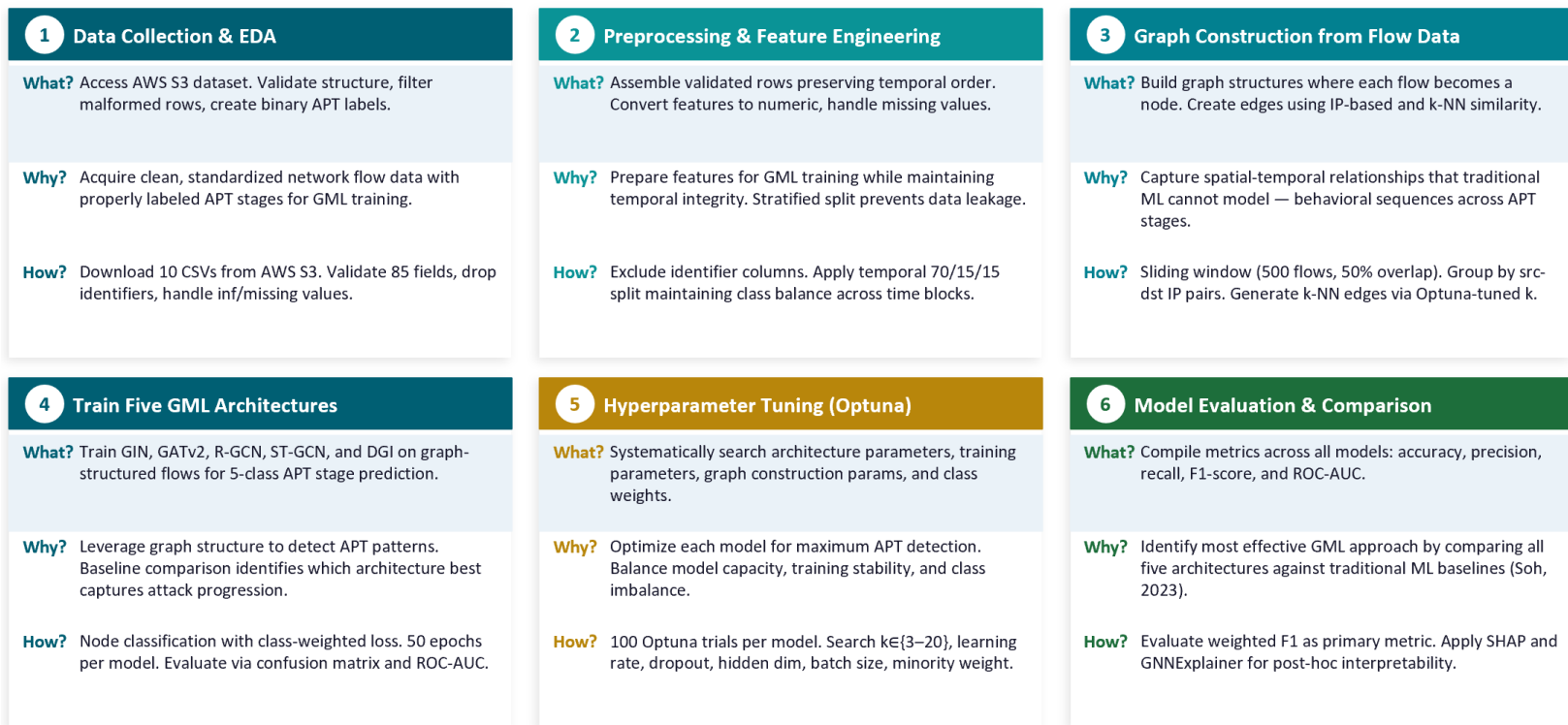


Figure C-1. Historical Graphical Model of Research from the original GWU GML praxis. Reproduced unchanged from the author-provided reference; it is not the current study's methodology. The full-resolution image is included with the supplement.

The new study retains the original convention of answering What, Why and How for each research stage. It replaces the old graph-training sequence with source qualification, matched evaluation, implemented comparison models, error-destination accounting, sensitivity analysis and verified reporting. This preserves the explanatory purpose of a GMR while keeping the current methodology faithful to the completed evidence.

Appendix D: Full-Release Evidence and Reproduction

This appendix reports the new uncapped extension separately from Appendix A's historical experiment inventory. The fixed class order is Benign, OtherAttackStage, MovementAuthorLabel and ExfiltrationAuthorLabel. All counts below are after duplicate handling and temporal-boundary exclusions.

D.1 net1011x

Table D-1. net1011x class support by split.

Split	Benign	Other	Movement	Exfiltration
train	1466865	0	0	0
calibration	5527	0	0	0
test	320359	0	0	0

current test results. Benign false-alert rate: 0.00%.

Table D-2. net1011x current test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	320359	100.00%	Unsupported	N/A
Other	0	Unsupported	Unsupported	0
Movement	0	Unsupported	Unsupported	0
Exfiltration	0	Unsupported	Unsupported	0

current roles test results. Benign false-alert rate: 0.00%.

Table D-3. net1011x current roles test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	320359	100.00%	Unsupported	N/A
Other	0	Unsupported	Unsupported	0
Movement	0	Unsupported	Unsupported	0
Exfiltration	0	Unsupported	Unsupported	0

D.2 net1012x

Table D-4. net1012x class support by split.

Split	Benign	Other	Movement	Exfiltration
train	597531	0	0	0
calibration	5132	0	0	0
test	131664	0	0	0

current test results. Benign false-alert rate: 0.00%.

Table D-5. net1012x current test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	131664	100.00%	Unsupported	N/A
Other	0	Unsupported	Unsupported	0
Movement	0	Unsupported	Unsupported	0
Exfiltration	0	Unsupported	Unsupported	0

current roles test results. Benign false-alert rate: 0.00%.

Table D-6. net1012x current roles test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	131664	100.00%	Unsupported	N/A
Other	0	Unsupported	Unsupported	0
Movement	0	Unsupported	Unsupported	0
Exfiltration	0	Unsupported	Unsupported	0

D.3 net1013x

Table D-7. net1013x class support by split.

Split	Benign	Other	Movement	Exfiltration
train	837124	12362	129	1740
calibration	5966	1811	0	906
test	195152	13270	35	3866

current test results. Benign false-alert rate: 0.01%.

Table D-8. net1013x current test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	195152	99.99%	Unsupported	N/A
Other	13270	97.23%	99.63%	49
Movement	35	20.00%	20.00%	28
Exfiltration	3866	71.00%	71.03%	1120

current roles test results. Benign false-alert rate: 0.01%.

Table D-9. net1013x current roles test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	195152	99.99%	Unsupported	N/A
Other	13270	97.22%	99.63%	49
Movement	35	20.00%	20.00%	28
Exfiltration	3866	71.03%	71.06%	1119

D.4 net1014x

Table D-10. net1014x class support by split.

Split	Benign	Other	Movement	Exfiltration
train	163270	0	0	0
calibration	2960	6	0	0
test	41193	2942	0	0

current test results. Benign false-alert rate: 0.00%.

Table D-11. net1014x current test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	41193	100.00%	Unsupported	N/A
Other	2942	0.00%	0.00%	2942
Movement	0	Unsupported	Unsupported	0
Exfiltration	0	Unsupported	Unsupported	0

current roles test results. Benign false-alert rate: 0.00%.

Table D-12. net1014x current roles test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	41193	100.00%	Unsupported	N/A
Other	2942	0.00%	0.00%	2942
Movement	0	Unsupported	Unsupported	0
Exfiltration	0	Unsupported	Unsupported	0

D.5 net1015x

Table D-13. net1015x class support by split.

Split	Benign	Other	Movement	Exfiltration
train	144089	0	129	0
calibration	2790	0	0	0
test	27619	0	24	1003

current test results. Benign false-alert rate: 0.04%.

Table D-14. net1015x current test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	27619	99.96%	Unsupported	N/A
Other	0	Unsupported	Unsupported	0
Movement	24	29.17%	29.17%	17
Exfiltration	1003	0.00%	0.00%	1003

current roles test results. Benign false-alert rate: 0.05%.

Table D-15. net1015x current roles test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	27619	99.95%	Unsupported	N/A
Other	0	Unsupported	Unsupported	0
Movement	24	29.17%	29.17%	17
Exfiltration	1003	0.00%	0.00%	1003

D.6 netgw

Table D-16. netgw class support by split.

Split	Benign	Other	Movement	Exfiltration
train	2116400	710	13333	6
calibration	8954	4461	886	0
test	707291	26712	12906	0

current test results. Benign false-alert rate: 0.30%.

Table D-17. netgw current test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	707291	99.70%	Unsupported	N/A
Other	26712	4.78%	4.80%	25429
Movement	12906	99.94%	99.94%	8
Exfiltration	0	Unsupported	Unsupported	0

current roles test results. Benign false-alert rate: 0.30%.

Table D-18. netgw current roles test outcomes.

Author-stage group	Support	Exact recall	Warning recall	Missed warnings
Benign	707291	99.70%	Unsupported	N/A
Other	26712	4.78%	4.80%	25430
Movement	12906	99.94%	99.94%	8
Exfiltration	0	Unsupported	Unsupported	0

D.7 Artifact Layout and Execution

The evidence package contains `FULL_RELEASE_PROTOCOL.json`, `FULL_RELEASE_RESULTS.json`, `FULL_RELEASE_AUDIT.json`, per-sensor `PREPARATION.json` and `RESULTS.json`, `full_release.py`, `audit_full_release.py`, the revised manuscript source and figure assets, and the original evidence archive. The local reproducibility bundle additionally contains saved models and every new calibration/test probability vector, target and row key. The source inventory records the original released flow-file hashes and acquisition provenance. Raw flow CSVs remain in the pinned local source directory and are not duplicated into the manuscript bundle.

Run `full_release.py` with Python, NumPy, scikit-learn, LightGBM and joblib using the recorded environment. The script's `ROOT` and `PRIVATE` paths locate public outputs and local prepared arrays. It requires the inventory at the recorded path and the matching raw source directory. Run `audit_full_release.py` after fitting to recompute count-based metrics independently from saved predictions. Dependencies and environment versions are captured in the final manifest. Repointing paths for another machine does not change the frozen analytical choices; verify source hashes before fitting.

The original 143-fit inventory and the extension's 12 fitting jobs are kept separate. A one-class LightGBM fitting job is identified as such and supplies a constant-class control; it is not counted as a successfully learned multiclass detector. The evidence is computationally reviewable and remains subject to source-label, campaign-dependence and institutional-review limitations.

Appendix E: Complete AIT Validation and Reproduction

E.1 Population, contract and audit

This appendix preserves the separate AIT experiment described in Sections 3.12 and 4.11. All eight archives are included in the reproduction package, with publisher MD5 verification, local SHA256 hashes, ZIP CRC checks and CC BY 4.0 attribution. Every eligible row enters its assigned final partition. Class support is shown below; the complete source accounting is in Table 3-10. Seeds share these same populations.

Table E-1. AIT eligible class support by execution.

Execution	Use	Benign	Other attack	Exfiltration
santos	Train	269,751	5,937	12,848
fox	Train	280,729	56,505	14,668
wardbeck	Train	370,045	5,099	15,076
russellmitchell	Train	206,674	155,250	15,778
shaw	Train	389,343	2,357	60,898
wheeler	Train	423,630	93,393	19,177
wilson	Test	568,013	40,321	21,219
harrison	Test	337,306	76,895	23,457

E.2 Seed 8101: Every Policy and Execution

Table E-2. Complete AIT outcomes for seed 8101.

Rows within each execution are paired across policies. Pooled rows combine Wilson and Harrison for the same seed.

Execution / policy	Macro-F1	Warnings	Missed exfil	Benign alerts	Acquire
wilson / Current	0.993388	99.967%	7	64	0.000%

Execution / policy	Macro-F1	Warnings	Missed exfil	Benign alerts	Acquire
wilson / Always history	0.994904	99.962%	8	53	100.000%
wilson / Entropy	0.994813	99.962%	8	84	67.647%
wilson / Error-focused	0.993765	99.967%	7	48	0.059%
harrison / Current	0.913033	99.437%	132	13746	0.000%
harrison / Always history	0.993611	99.599%	94	686	100.000%
harrison / Entropy	0.993542	99.599%	94	715	60.727%
harrison / Error-focused	0.913152	99.442%	131	13744	1.454%
pooled / Current	0.947968	99.689%	139	13810	0.000%
pooled / Always history	0.994715	99.772%	102	739	100.000%
pooled / Entropy	0.994640	99.772%	102	799	64.809%
pooled / Error-focused	0.948175	99.691%	138	13792	0.631%

E.3 Seed 8102: Every Policy and Execution

Table E-3. Complete AIT outcomes for seed 8102.

Rows within each execution are paired across policies. Pooled rows combine Wilson and Harrison for the same seed.

Execution / policy	Macro-F1	Warnings	Missed exfil	Benign alerts	Acquire
wilson / Current	0.993433	100.000%	0	45	0.000%
wilson / Always history	0.994895	99.986%	3	52	100.000%
wilson / Entropy	0.994772	99.986%	3	83	84.861%

Execution / policy	Macro-F1	Warnings	Missed exfil	Benign alerts	Acquire
wilson / Error-focused	0.993790	100.000%	0	47	3.195%
harrison / Current	0.907972	99.442%	131	14815	0.000%
harrison / Always history	0.989431	99.582%	98	1255	100.000%
harrison / Entropy	0.989368	99.582%	98	1282	77.170%
harrison / Error-focused	0.908027	99.531%	110	14866	7.457%
pooled / Current	0.944796	99.707%	131	14860	0.000%
pooled / Always history	0.992528	99.774%	101	1307	100.000%
pooled / Entropy	0.992445	99.774%	101	1365	81.707%
pooled / Error-focused	0.944940	99.754%	110	14913	4.943%

E.4 Seed 8103: Every Policy and Execution

Table E-4. Complete AIT outcomes for seed 8103.

Rows within each execution are paired across policies. Pooled rows combine Wilson and Harrison for the same seed.

Execution / policy	Macro-F1	Warnings	Missed exfil	Benign alerts	Acquire
wilson / Current	0.993449	100.000%	0	58	0.000%
wilson / Always history	0.994918	99.962%	8	53	100.000%
wilson / Entropy	0.994817	99.962%	8	80	78.991%
wilson / Error-focused	0.993479	100.000%	0	48	0.033%
harrison / Current	0.908889	99.424%	135	14618	0.000%

Execution / policy	Macro-F1	Warnings	Missed exfil	Benign alerts	Acquire
harrison / Always history	0.993993	99.595%	95	642	100.000%
harrison / Entropy	0.993902	99.595%	95	670	73.368%
harrison / Error-focused	0.909205	99.497%	118	14577	4.282%
pooled / Current	0.945386	99.698%	135	14676	0.000%
pooled / Always history	0.994915	99.769%	103	695	100.000%
pooled / Entropy	0.994824	99.769%	103	750	76.685%
pooled / Error-focused	0.945585	99.736%	118	14625	1.775%

E.5 Reproduction Materials

Campaign_Validation_Full_Evidence.zip contains the full eight-archive release, eight prepared arrays, 36 classifier objects, six selector objects, all development out-of-fold predictions and every held-out probability/action array. Reports include ACQUISITION.json, QUALIFICATION.json, PREPARATION.json, PROTOCOL.md, FREEZE.json, FIT_LOG.json, RESULTS.json, RESULTS.csv, SUMMARY.json, AUDIT.json, ARTIFACTS.json, NATIVE_LABEL_DIAGNOSTIC.json and SELECTOR_DIAGNOSTIC.json. The latter diagnostic is descriptive after fitting and does not alter the frozen policies. Code and pinned compute dependencies are included. The computational audit passes 1,245 checks.

The local data root is C:/w/campaign_validation_20260928. Run acquire_qualify.py, validate.py prepare, validate.py run, and audit.py with the recorded environment. Reprint documented absolute paths when moving machines and record the

path amendment. The complete output bundle accompanies the integrated paper and defense. Historical and full-release UNRAVELED evidence retain their own source bindings and experimental contracts; their metrics are not pooled with this different task.